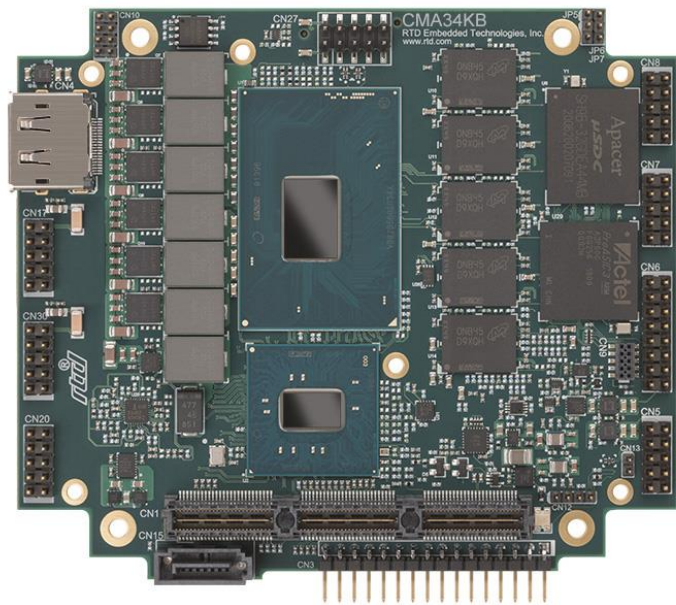
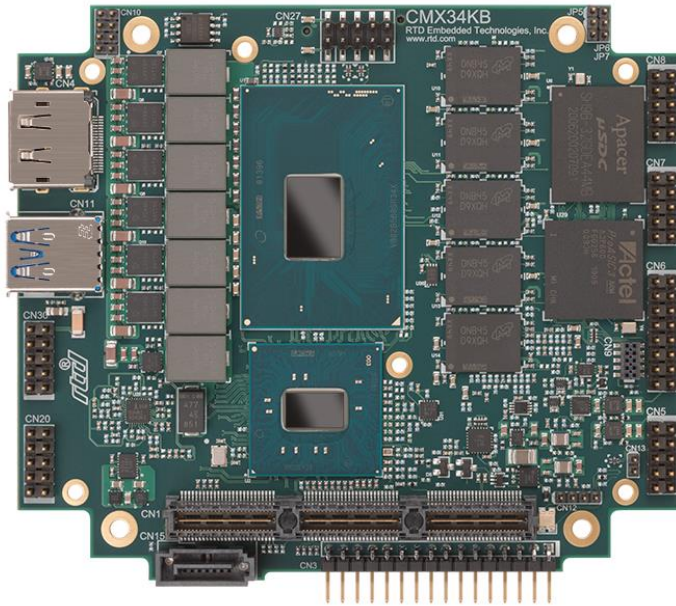


CMX34KB / CMA34KB



User's Manual

BDM-610000089 Revision E



CMX34KB/CMA34KB cpuModules™ User's Manual

RTD Document Number: BDM-610000089 Revision E

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ISO9001 and AS9100



Revision History

Revision	Date	Reason for Change
A	9/7/2021	Initial release
B	5/15/2024	Corrected CN3 connector description and mating connector model. Added CMA34KB version of cpuModule
C	6/18/2024	Added table for USB2.0 pinout
D	8/21/2024	Fixed Table 24 signals
E	9/16/2024	Fixed Figure 6 (CN17) description

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Appendix E Limited Warranty

Chapter 1 Introduction

This manual provides comprehensive hardware and software information for users developing with the CMX34KB/CMA34KB PCIe/104 cpuModule.



Note Read the specifications beginning on page 21 prior to designing with the cpuModule.

This manual is organized as follows:

- | | |
|-------------------|--|
| Chapter 1 | <i>Introduction</i>
introduces main features and specifications |
| Chapter 2 | <i>Getting Started</i>
provides abbreviated instructions to get started quickly |
| Chapter 3 | <i>Connecting the cpuModule</i>
provides information on connecting the cpuModule to peripherals |
| Chapter 4 | <i>Using the cpuModule</i>
provides information to develop applications for the cpuModule, including general cpuModule information, detailed information on storing both applications and system functions, and using utility programs |
| Appendix A | <i>Hardware Reference</i>
lists jumper locations and settings, physical dimensions, and processor thermal management |
| Appendix B | <i>Troubleshooting</i>
offers advice on debugging problems with your system |
| Appendix C | <i>IDAN® Dimensions and Pinout</i>
provides connector pinouts for the cpuModule installed in an RTD Intelligent Data Acquisition Node (IDAN) frame |
| Appendix D | <i>Additional Information</i>
lists sources and websites to support the cpuModule installation and configuration |
| Appendix E | <i>Limited Warranty</i> |

CMX34KB/CMA34KB cpuModules

RTD's CMX34KB/CMA34KB cpuModule represents the latest in high performance PCIe/104 embedded controllers with a Type 2 bus. This design is based on an Intel Kaby Lake Xeon processor with CM238 Platform Controller Hub. The CMX34KB/CMA34KB is offered with either a 2.2Ghz Quad-Core processor or a 3GHz Quad-Core processor.

Gen 3.0 PCIe support consists of 8 PCIe x1 lanes and 4 PCIe x4 lanes. The top PCIe bus connector contains 4 x1 lanes and 2 x4 lanes. The bottom connector also contains 4 x1 lanes and 2 x4 lanes.

Onboard memory consists of 16 GB of embedded ECC DDR4 DRAM with a speed of 2400 MT/s (1200 MHz).

The cpuModule's DisplayPort video interface supports audio and a maximum resolution of 4096 x 2304.

A Serial-ATA (SATA) controller provides six SATA 3.0 links. Two links are to the top and two links to the bottom PCIe connectors. One of the remaining two links is routed to an embedded 60 GB drive and the last link is routed to an onboard SATA connector.

These cpuModules include an xHCI controller available in two configurations. The CMA34KB has two 10-pin DIL headers for a total of four USB 2.0 ports. The CMX34KB has one 10-pin DIL header for two USB 2.0 ports and one USB Type A receptacle for one USB 3.0 port. Both the CMX34KB and CMA34KB cpuModules have an additional USB 2.0 port routed to the Utility Port for use of a keyboard and mouse.

Network connectivity is provided by two 10/100/1000 Mbps Intel i210 Ethernet controllers.

Other features include RS-232/422/485 COM ports and Advanced Digital I/O (aDIO).

RTD has implemented additional features for increased utility and security. These include RS-232/422/485 COM ports, Advanced Digital I/O (aDIO), an Advanced Watchdog Timer and a Trusted Platform Module (TPM 2.0). The CMX34KB/CMA34KB is also made available in a rugged and fan-less IDAN enclosure.

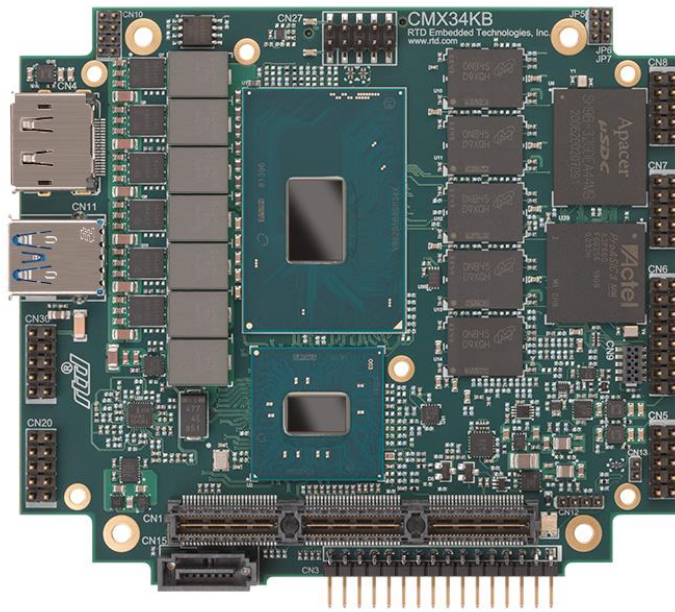


Figure 1 CMX34KB cpuModule (top view)

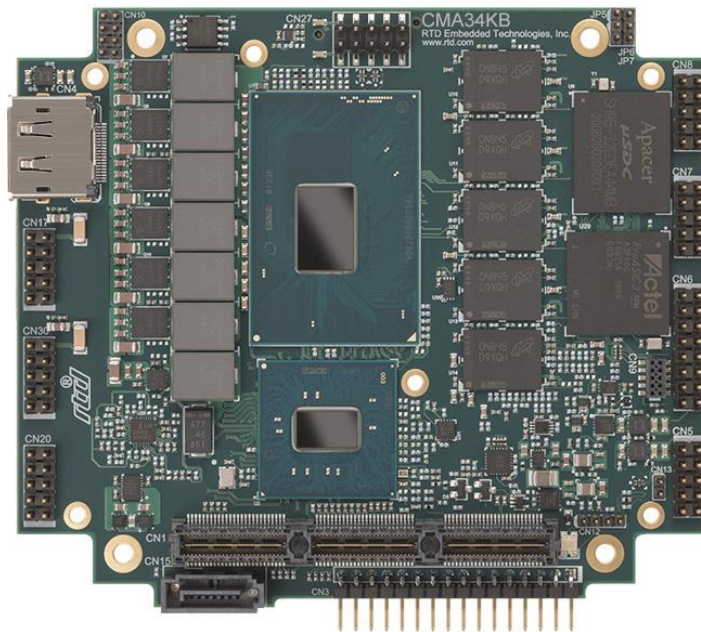


Figure 2 CMA34KB cpuModule (top view)

You can easily customize the cpuModule by stacking PCIe/104 modules such as video controllers, Digital Signal Processors, drive carriers, LAN controllers, or analog and digital data acquisition modules. Stacking modules onto the cpuModule avoids expensive installations of backplanes and card cages and preserves the module's compactness.

The cpuModule uses the RTD Enhanced BIOS. Functionality in the BIOS allows booting from hard disk, UEFishell, USB, or network booting.

The cpuModule and BIOS are compatible with any real-time operating systems for PC compatible computers, although these may require creation of custom drivers to use the aDIO and watchdog timer.

aDIO

RTD's exclusive aDIO™ is 12 digital bits configured as 8 bit-direction programmable and 4-bit port-direction programmable I/O, plus 2 strobe inputs giving you any combination of inputs and outputs. Match, event, and strobe interrupt modes mean no more wasting valuable processor time polling digital inputs. Interrupts are generated when the 8 bit-direction programmable digital inputs match a pattern or on any value change event. Bit masking allows selecting any subgroup of eight bits. The strobe input latches data into the bit-programmable port and generates an interrupt.

Ordering Information

The CMX34KB/CMA34KB cpuModule is available with a selection of processors and heatsinks. The cpuModule can also be purchased as part of an Intelligent Data Acquisition Node (IDAN[®]) building block, which consists of the cpuModule and a milled aluminum IDAN frame. The IDAN building block can be used in just about any combination with other IDAN building blocks to create a simple but rugged PC/104 stack. Refer to Appendix C, *IDAN[®] Dimensions and Pinout*, for more information. The CMX34KB/CMA34KB cpuModule can also be purchased as part of a custom-built RTD HiDAN[®] or HiDAN^{plus} High Reliability Intelligent Data Acquisition Node. Contact RTD for more information on its high reliability PC/104 systems.

CMX34KB/CMA34KB Model Options

The basic cpuModule model options are shown below. Refer to the RTD website (www.rtd.com) for more detailed ordering information and any new variations that may be available. RTD also offers customer custom modifications. Contact RTD Sales for details.

Table 1 CMX34KB/CMA34KB cpuModule Model Options¹

Part Number	Description	Stack Height
CMX34KBQ2200HR-16384/S60GX	Intel Xeon [®] E3-1505L, Quad Core 2.2 GHz 16GB DDR4-SDRAM, USB3.0 via Type A Receptacle	22mm
CMX34KBQ3000HR-16384/S60GX	Intel Xeon [®] E3-1505M, Quad Core 3.0 GHz 16GB DDR4-SDRAM, USB3.0 via Type A Receptacle	22mm
CMX34KBD2100HR-16384/S60GX	Intel i3-7102E, Dual Core 2.1 GHz 16GB DDR4-SDRAM, USB3.0 via Type A Receptacle	22mm
CMA34KBQ2200HR-16384/S60GX	Intel Xeon [®] E3-1505L, Quad Core 2.2 GHz 16GB DDR4-SDRAM, USB2.0 via 10-pin DIL	22mm
CMA34KBQ3000HR-16384/S60GX	Intel Xeon [®] E3-1505M, Quad Core 3.0 GHz 16GB DDR4-SDRAM, USB2.0 via 10-pin DIL	22mm
CMA34KBD2100HR-16384/S60GX	Intel i3-7102E, Dual Core 2.1 GHz 16GB DDR4-SDRAM, USB2.0 via 10-pin DIL	22mm
CMX34KBD2100HR-16384/S60GXF	CMX34KBD2100HR-16384/S60GX with Flat Heatspreader	22mm
CMX34KBQ2200HR-16384/S60GXF	CMX34KBQ2200HR-16384/S60GX with Flat Heatspreader	22mm
CMX34KBQ3000HR-16384/S60GXF	CMX34KBQ3000HR-16384/S60GX with Flat Heatspreader	22mm
CMA34KBD2100HR-16384/S60GXF	CMA34KBD2100HR-16384/S60GX with Flat Heatspreader	22mm
CMA34KBQ2200HR-16384/S60GXF	CMA34KBQ2200HR-16384/S60GX with Flat Heatspreader	22mm
CMA34KBQ3000HR-16384/S60GXF	CMA34KBQ3000HR-16384/S60GX with Flat Heatspreader	22mm

1. All ordering options include a Trusted Platform Module (TPM 2.0).

Difference Between CMX34KB and CMA34KB

- The difference in USB connectors on these models is on the left side of the boards, just below the DisplayPort receptacle.
- The CMA34KB and IDAN-CMA34KB have a 10-pin DIL header for USB 2.0, CN17.
- The CMX34KB has a USB Type A receptacle for USB 3.0, CN11.

Cable Kits and Accessories

For maximum flexibility, RTD does not provide cables with the cpuModule. You may wish to purchase the CMX34KB/CMA34KB cpuModule cable kit (XK-CM125), which contains:

- Utility Port 2.0 multi-function cable (2x USB 2.0 ports, battery, reset button, power button, speaker)
- Two serial port cables (DIL-10 to DSUB-9)
- aDIO cable (DIL-16 to DSUB-25)
- Two USB cables (5-pin SIL to USB A)
- PCIe/104 Type 2 break-out board (provides standard PC interfaces for SATA and USB devices)
- Two ethernet cables (DIL-10 to RJ-45)

For additional accessories, refer to the RTD website.

Board Features

- Intel Xeon® E3-1505 Series Processor:

Part Number	Cores	Core Frequency	L1 Cache	L2 Cache	L3 Cache	Stack Height
CMX34KBD2100	Two	2.1 GHz	128 KByte	1 MByte	8 MByte	22mm
CMX34KBQ2200	Four	2.2 GHz	128 KByte	1 MByte	8 MByte	22mm
CMX34KBQ3000	Four	3.0 GHz	128 KByte	1 MByte	8 MByte	22mm
CMA34KBD2100	Two	2.1 GHz	128 KByte	1 MByte	8 MByte	22mm
CMA34KBQ2200	Four	2.2 GHz	128 KByte	1 MByte	8 MByte	22mm
CMA34KBQ3000	Four	3.0 GHz	128 KByte	1 MByte	8 MByte	22mm

- Discreet L1 and L2 cache structures for each core.
 - Intel Xeon® technology
 - i3-7102E Dual Core Processor
 - E3-1505L Quad Core Processor
 - E3-1505M Quad Core Processor
 - Thermal Throttling reduces clock speed to prevent thermal runaway
 - 14 nm process
 - Max Core Temperature of 100 C
- 16 Gigabytes BGA DDR4 SDRAM with ECC
 - Dual-channel memory interface
 - Transfer rates up to 2400 MT/s Data Rate
 - Surface Mounted for maximum reliability
- Stackable 156-pin PCIe/104 Type 2 bus on top
 - Two PCI Express x4 Links. Gen 3, 8 GT/s
 - Four PCI Express x1 Links. Gen 3, 8 GT/s
 - Two SATA 3.0, Rev 3.0 600MB/s
 - ATX Power Supply Signaling
 - RTC Battery
- Stackable 156-pin PCIe/104 Type 2 bus on bottom
 - Two PCI Express x4 Links. Gen 3, 8 GT/s
 - Four PCI Express x1 Links. Gen 3, 8 GT/s
 - Two SATA 3.0, Rev 3.0 600MB/s
 - Two USB 2.0
 - ATX Power Supply Signaling
 - RTC Battery
- Advanced Thermal Management
 - Thermal Monitor throttles processor and memory to prevent thermal runaway
 - Modified heatsinks and flat heat spreaders are available
 - Mini Fan Heatsink with Auto Fan control
 - Passive Structural Heatsink & Heat Pipes in IDAN and HiDAN System Configurations

- Advanced Configuration and Power Interface (ACPI)
 - Wake events include:
 - USB event (device insertion, keyboard keystroke, etc.)
 - Power Switch
- Real-Time Clock (external battery required to maintain time only)
- Nonvolatile storage of CMOS settings without battery
- Advanced Watchdog timer
- Complete PC-compatible Single Board Computer
- Trusted Platform Module (TPM)
 - Fully compliant to the Trusted Computing Group (TCG) Trusted Platform Module (TPM) version 2.0 specification
 - Provides for secure storage of cryptographic keys

I/O

- DisplayPort Interface Rev. 1.4
 - Max Resolution 4096 x 2304
 - Audio over DisplayPort
- Two Gigabit Ethernet Ports
 - Intel WGI210IT PCI Express x1 Controller
 - 10/100/1000 Auto-negotiation
 - Jumbo Frame Support (9.5 kB)
 - Automatic MDI/MDI-X crossover capable
- Software-configurable RS-232/422/485 serial ports
 - 16550 compatible UARTs for high-speed
 - 120 Ohm Termination resistors for RS-422/485 through BIOS Configuration
 - Fully jumperless configuration
 - Increased FIFO size of 512 bytes for both transmit and receive.
 - Supports echo cancellation, Auto-RTS and Auto-CTS
 - Each serial port can be configured as two truncated serial ports for a total of four serial ports
- Advanced Digital I/O (aDIO)
 - One 8-bit bit-programmable I/O with Advanced Digital Interrupt Modes
 - One 4-bit port programmable as input or output
 - Event Mode Interrupt generates an interrupt when any input bit changes
 - Match Mode Interrupt generates an interrupt when input bits match a preset value
 - External Strobe Mode latches 8 data inputs and generates an interrupt
 - Two Strobes can be configured as readable inputs
- USB 2.0 (Universal Serial Bus) Ports
 - Supports 480 Mb/s (high-speed), 12Mb/s (Full Speed), and 1.5Mbps (low speed) peripherals
 - Automatic resettable over-current protection (500 mA @ 5 Vdc per port)
 - USB Boot capability
 - Three ports on 0.1" I/O headers: two on CN27, and one on CN5
 - The CMA34KB has two additional USB 2.0 ports at CN17. The CMX34KB has one USB 3.0 port at CN11.

- One USB 3.0 (Universal Serial Bus) Port on the CMX34KB
 - Supports 625 MByte/s (Super Speed)
 - USB Boot capability
 - On CN11, a USB A connector 
 - Automatic resettable over-current protection (750 mA @ 5 Vdc)
 - Not included on the CMA34KB
- Serial ATA (SATA)
 - Six SATA revision 3.0 links
 - Two links to the top-side PCIe connector
 - Two links to the bottom-side PCIe connector
 - One link to an onboard SATA connector CN15
 - One link to the onboard surface-mount SATA Flash drive
 - Onboard 60 GB SATA 3.0 Flash drive
 - Built in Wear Leveling, Error Correction and Bad Block Management
 - SMART supported (Self-Monitoring, Analysis and Reporting Technology)
 - Compatibility mode supports legacy operating systems.
- Utility Port 2.0 Connector
 - 1x USB 2.0 (Universal Serial Bus) Port with 500 mA over-current protection
 - Speaker port (0.1 W output)
 - Hardware Reset input
 - Soft Power Button input
- Battery input for Real Time Clock
- Power Management
 - ACPI - Advanced Configuration and Power interface
 - ATX support for “Soft Off”
 - ATX Power signals

RTD Enhanced BIOS

- User-configurable using built-in Setup program
- Flash-based backup of CMOS Settings; no battery required to store CMOS settings
- Supports boot from SATA, USB, or Network
- UEFI (Unified Extensible Firmware Interface) Shell
- Special RTD Enhancements

Block Diagram

The next figure shows a simplified block diagram of the CMX34KB cpuModule.

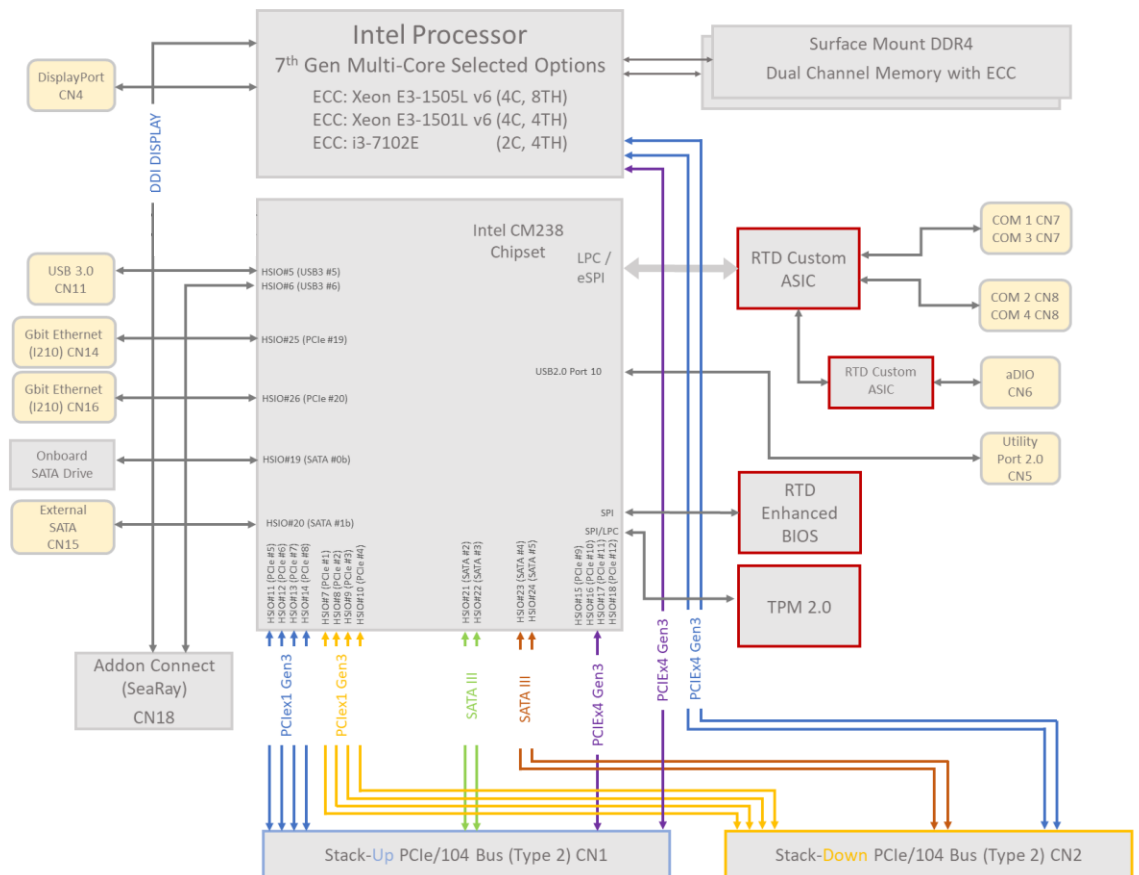


Figure 2 CMX34KB cpuModule Simplified Block Diagram

The following figure shows a simplified block diagram of the CMA34KB cpuModule.

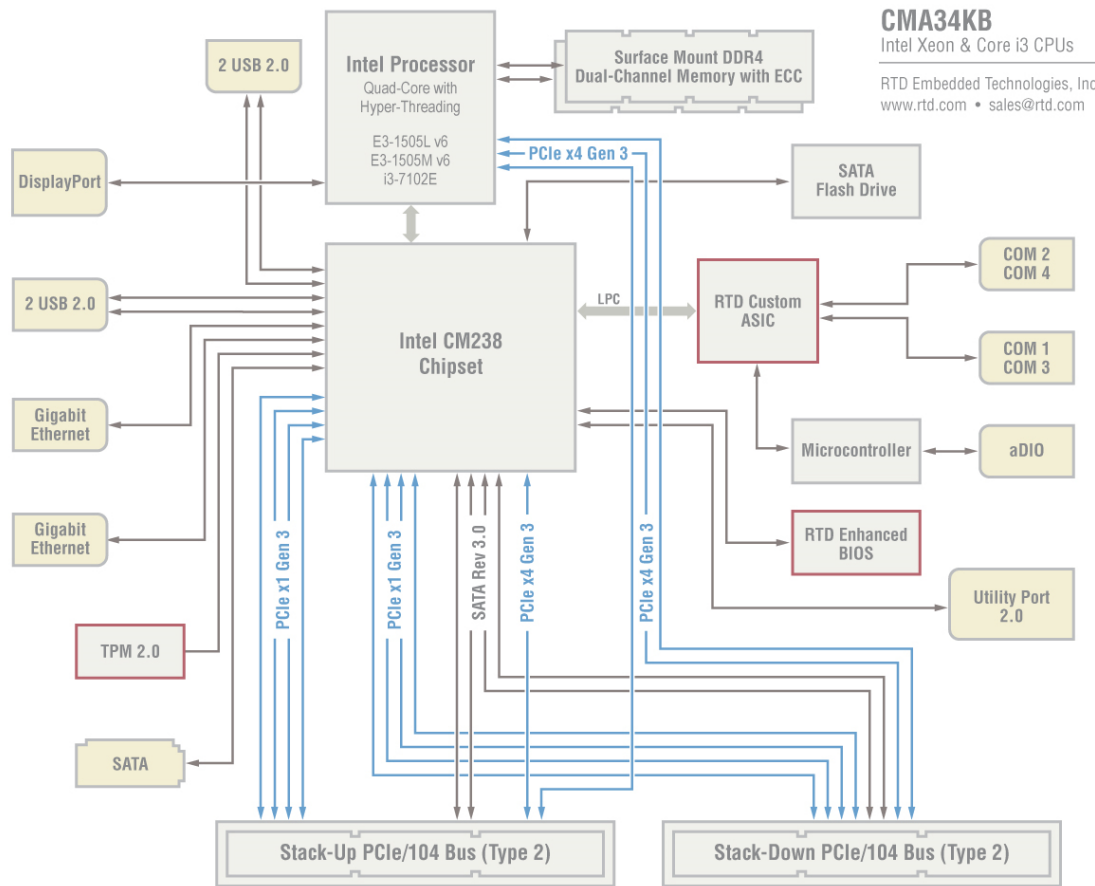


Figure 3 CMA34KB cpuModule Simplified Block Diagram

Specifications

Physical Characteristics

Basic dimensions and the weight of the CMX34KB/CMA34KB are listed below. For a more detailed dimensional drawing, refer to the *Physical Dimensions* section.

- Dimensions
 - Board Size (connectors not included)
 - Length (L): 99.695 mm L (3.925")
 - Width (W): 115.57 mm W (4.550")
 - Stand-off Height Above the CPU:
 - Quad-core models 22.000 mm (0.866")
 - Stand-off Height Below the CPU:
 - All versions: 15.240mm (0.600")
- Weight: Approximately 0.261 Kg (0.574 lbs.) with fan Heatsink
- Weight: Approximately 0.377 Kg (0.832 lbs.) with flat Heat spreader
- Weight: Approximately 0.140 Kg (0.308 lbs.) just board
- IDAN Weight: Approximately 0.765 Kg (1.686 lbs.)

Power Consumption

Exact power consumption depends on the actual application. Typical power consumption of the CMX34KB/CMA34KB is listed in Table 3. It is expected that power consumption of a typical application will be a combination of these scenarios.

Table 2 Typical Power Consumption

Part Number	Typical Power Consumption
CMX34KBQ2200HR-16384	21.7 W
CMX34KBQ3000HR-16384	34.8 W
CMX34KBD2100HR-16384	16.8 W
CMA34KBQ2200HR-16384	21.7 W
CMA34KBQ3000HR-16384	34.8 W
CMA34KBD2100HR-16384	16.8 W



Note The measurements listed above are intended to represent common usage scenarios. The *cpuModule*'s power consumption is heavily influenced by the running software and system activity.

Operating Conditions

Table 3 Operating Conditions

Symbol	Parameter	Test Condition	Min.	Max.
V_{CC5}	5V Supply Voltage		4.75V	5.25V
V_{CC12}	12V Supply Voltage		11.4V	12.6V
V_{CCSTBY}	5V Standby Voltage ¹		4.75V	5.25V
I_{CCSTBY}	5V Standby Current ¹		-	800mA
T_a	Ambient Operating Temperature ²		-40	+70C
T_s	Storage Temperature		-40	+70C
R_h	Humidity	Non-Condensing	0	90%
MTBF	Mean Time Before Failure (hours)	CMX34KBQ3000HR-16384/S60GX	344,416	
		CMX34KBQ2200HR-16384/S60GX	344,416	
		CMX34KBD2100HR-16384/S60GX	344,416	

1. With supplied heat sink solution. Depending on the CPU usage, performance may degrade as the ambient temperature approaches the maximum. Contact RTD Tech Support for more information.
2. Calculation Model: Telcordia Issue 2; Calculation Method: Method 1 Case 3; Temperature: +30C; Environment: Ground Benign, Controlled

Electrical Characteristics

The table below lists the Electrical Characteristics of the CMX34KB/CMA34KB. Operating outside of these parameters may cause permanent damage to the cpuModule.

Table 4 Electrical Characteristics

Symbol	Parameter	Test Condition	Min.	Max.
USB Ports				
I_{oc}	Overcurrent Limit	Each Port USB2.0	0.51 A	0.75 A
I_{oc}	Overcurrent Limit	Each Port USB3.0	0.75 A	1.06 A
DisplayPort				
I_{VCC}	Supply Current	—		500 mA
Serial Ports - RS-232				
V_{OH}	Output Voltage High	R _L = 3k	5.0 V	7.0 V
V_{OL}	Output Voltage Low	R _L = 3k	-7.0 V	-5.0 V
V_{IH}	Input Voltage High	—	2.0 V	15 V
V_{IL}	Input Voltage Low	—	-15 V	0.6 V
Serial Ports - RS-422/485				
V_{OD1}	Differential Output	R _L = 100 Ohm	2.0 V	
V_{OD2}	Differential Output	R _L = 54 Ohm	1.5 V	
V_{OC}	Common Mode Output	R _L = 54 or 100 Ohm		3.0 V
V_{TH}	Differential Input Threshold		-200 mV	-50 mV
V_I	Absolute Max Input Voltage		-18 V	-18 V

Table 5 Electrical Characteristics

Symbol	Parameter	Test Condition	Min.	Max.
aDIO²				
V_{OH}	Output Voltage High	I _{OH} = -6 mA	2.3 V	3.3 V
V_{OL}	Output Voltage Low	I _{OL} = 10 mA	0.0 V	0.6 V
V_{IH}	Input Voltage High	—	1.625 V	3.6 V
V_{IL}	Input Voltage Low	—	-0.3 V	0.495 V
I_{ADIOvcc}	Supply current	—		500 mA
T_{MIN}	Update Interval	—	60us	—
Utility Port 2.0 Connector (CN5)				
V_{RTC}	Input RTC Voltage ¹	—	2.5 V	3.6 V
I_{RTC}	RTC Battery Current	—		< 4 uA
I_{UTILvcc}	Utility Supply Current	—		350 mA

1. Date and time lost when power is completely removed from the system including the CMOS battery. Date and time are not required for board operation. BIOS will set default values on the next power cycle.

2. Default build is 3.3V signaling. Ask sales about 5V signaling option.

Migrating to RTD's Intel Xeon Series cpuModules

When migrating to RTD's Intel Xeon Series cpuModules, there are several differences from preceding RTD cpuModule product families (Chief River, Baytrail, Montevina) of which the customer should be aware.

Connector Pinout & Function Differences

The Intel Xeon Series cpuModules have several connector-related differences, which are summarized below. Complete information about the connectors on the CMX34KB/CMA34KB can be found in Chapter 3, *Connecting the cpuModule*.

New I/O Connectors on the CMX34KB

The CMX34KB cpuModule introduces a new I/O connector that was not present on some previous RTD cpuModules:

- **USB 3.0 Connector (CN11)**
 - Provides a USB 3.0 connection via a Type A receptacle. 
 - Note: The USB 3.0 connection is replaced with a USB 2.0 connection on the CMA34KB.

Note: The CMA34KB does not have a Type A receptacle. See Connector Differences below.

Connector Differences

While some connectors on the CMX34KB/CMA34KB have identical pinouts as on previous cpuModule generations, the features of the connector are sometimes different. Other connectors have entirely different pinouts than those on previous cpuModule generations. Connectors with pinout and feature differences include:

- **Auxiliary Power Connector (CN3)**
 - RTD Intel Xeon Series CPU **requires** +12V. May optionally be connected to CN3 if it is required by any other device in the system.
 - 12 pins (unlike the 10-pin connector on RTD Montevina cpuModules)
 - +5V and +12V inputs (unlike +5V only auxiliary power connectors found on Montevina cpuModule). Note: CMX34KB/CMA34KBs cpuModule only uses 12V from CN3.
 - The Auxiliary Power Connector excludes ATX power signaling. The ATX power signals are provided on the PCIe/104 Type 2 connector.

CMA34KB

- **USB 2.0 (CN17)**
 - The CMA34KB and IDAN-CMA34KB have a 10-pin DIL header for USB 2.0, **CN17**.
 - The CMX34KB do not have the 10-pin header. It has been replaced by the USB Type A receptacle, **CN11**.

Software Differences

The Intel Xeon Series cpuModules have several software-related differences, which are summarized below. Complete information about these CMX34KB/CMA34KBs features can be found in Chapter 4, *Using the cpuModule*.

aDIO

While the 16-pin aDIO connector is pin-for-pin compatible with previous generations of RTD cpuModules, the hardware is different, and requires a 60-microsecond delay for a read or write cycle to complete.

UEFI Shell

Previous generations of RTD cpuModules contained a Failsafe Boot ROM image, which permitted the cpuModule to boot to a DOS prompt even when no disk drives were connected to the system. The RTD Xeon Series cpuModules provide a similar prompt, called the Unified Extensible Firmware Interface (UEFI) Shell, which provides an environment to execute simple commands when no bootable devices are connected to the system. EFI Shell commands can easily be found on the internet.

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Chapter 2 Getting Started

For many users, the factory configuration of the CMX34KB/CMA34KB cpuModule can be used to get a PC/104 system operational. You can get your system up and running quickly by following the simple steps described in this chapter, which are:

1. Before connecting the cpuModule, the user must be properly grounded to prevent electrostatic discharge (ESD). For more information, refer to *Proper Grounding Techniques* .
2. Connect the Utility Port 2.0 cable.
3. Connect a USB keyboard.
4. Connect a monitor to the DisplayPort connector or DisplayPort interface.
5. Connect power.
6. Default BIOS configuration.
7. UEFI (Unified Extensible Firmware Interface) shell.

Refer to the remainder of this chapter for details on each of these steps.

Connector Locations

Figure 3 and Figure 4 show the connectors of the CMX34KB and CMA34KB cpuModules, respectively.

Figure 3 CMX34KB cpuModule (top view)

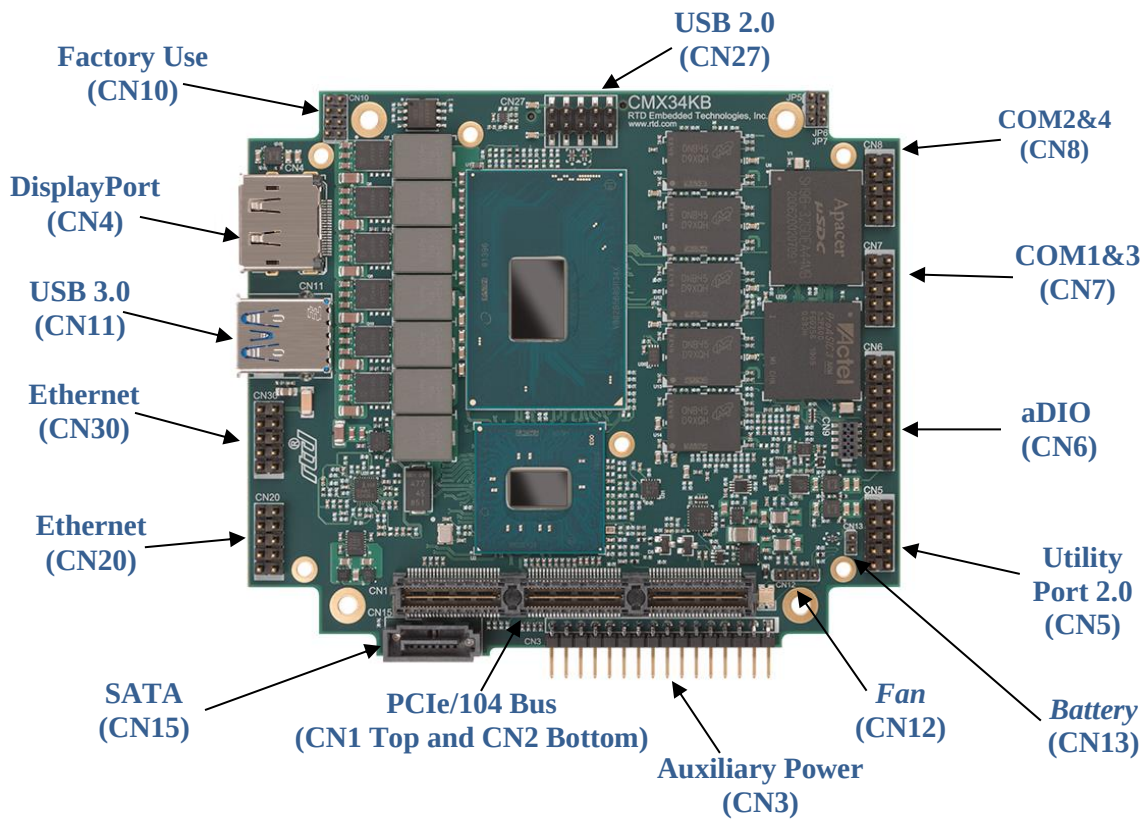
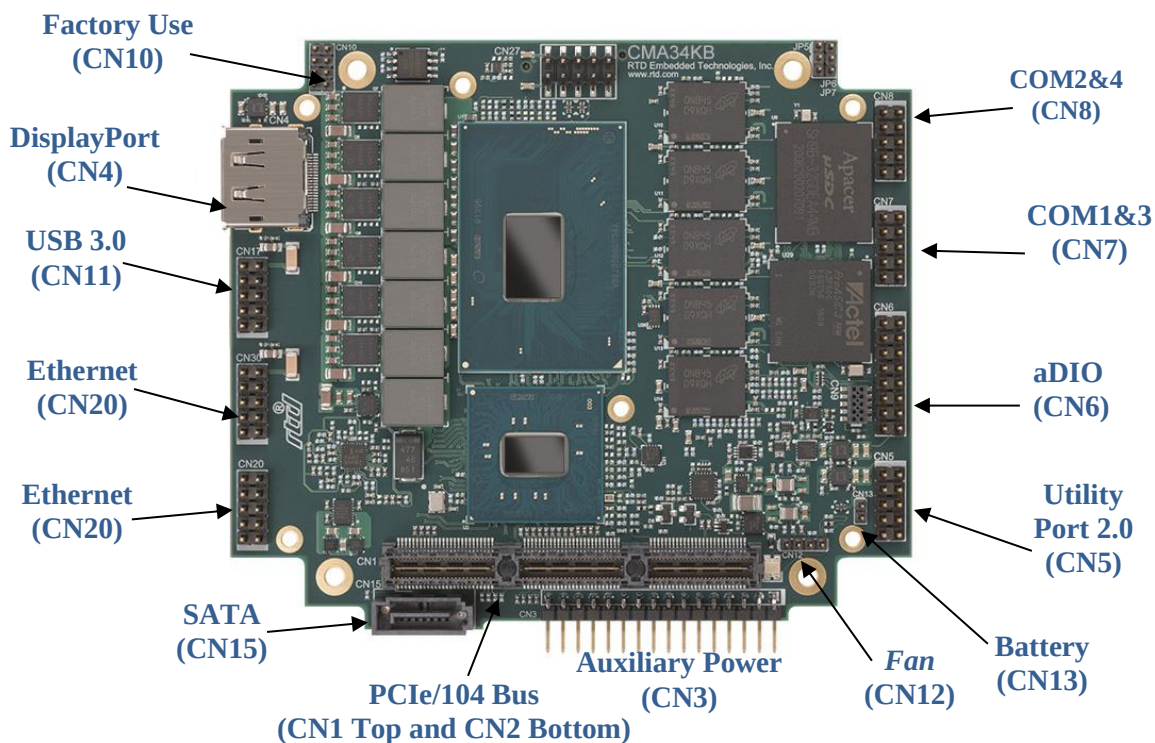


Figure 4 CMA34KB cpuModule (top view)



Selecting the Stack Order for the CMX34KB/CMA34KB

There are several things to consider when selecting the order of boards in the stack. Before selecting the order, be sure to determine which bus connector on each board is the “Active” bus. Typically, if a peripheral module has both PCIe and PCI bus connectors, only the PCIe is active, and the PCI is pass-through. The following is a list of rules to use to determine the stack order:

1. The PCIe connectors above and below the CPU have separate bus signals that are not electrically connected. Therefore, it is possible to attach boards to the PCIe connector above and below the CPU.
2. Any board that uses a SATA link on one of the PCIe connectors must be within one board of the CPU.
3. Generally, keep higher speed boards closer to the CPU.
4. If using lane repopulation, consume all lanes before making the last lane repopulate, to preserve bandwidth.
5. Keep the power supply as close as possible to the CPU due to the high instantaneous current demands of the CPU.
6. If possible, keep a PCIe spacer directly above the CPU to maximize available cool air to the heat sink.
7. If a PCI bus is needed, a maximum of four PCI boards may be attached to any PCI bus. If more than four PCI boards are needed in a system, a PCIe to PCI bridge is required.
8. If a PCI bus is needed, the PCIe-to-PCI bridge must be at the extreme end of the PCI bus segment, and all the peripheral cards at the other end. There may be up to eight PCI pass-through connectors between the PCIe to PCI bridge and the peripheral cards.
9. If a PCI bus is needed, there must be no more than two boards between the first PCI peripheral and the last PCI peripheral. If there are four PCI peripheral cards in a PCI bus segment, there may not be any PCI pass-through connections between them.

Stack Example

The figure below shows an example of a complete system stack. Most systems will be a subset of this example. This example stack may be further expanded with PCIe-to-PCIe bridges or a PCIe to PCI bridge.

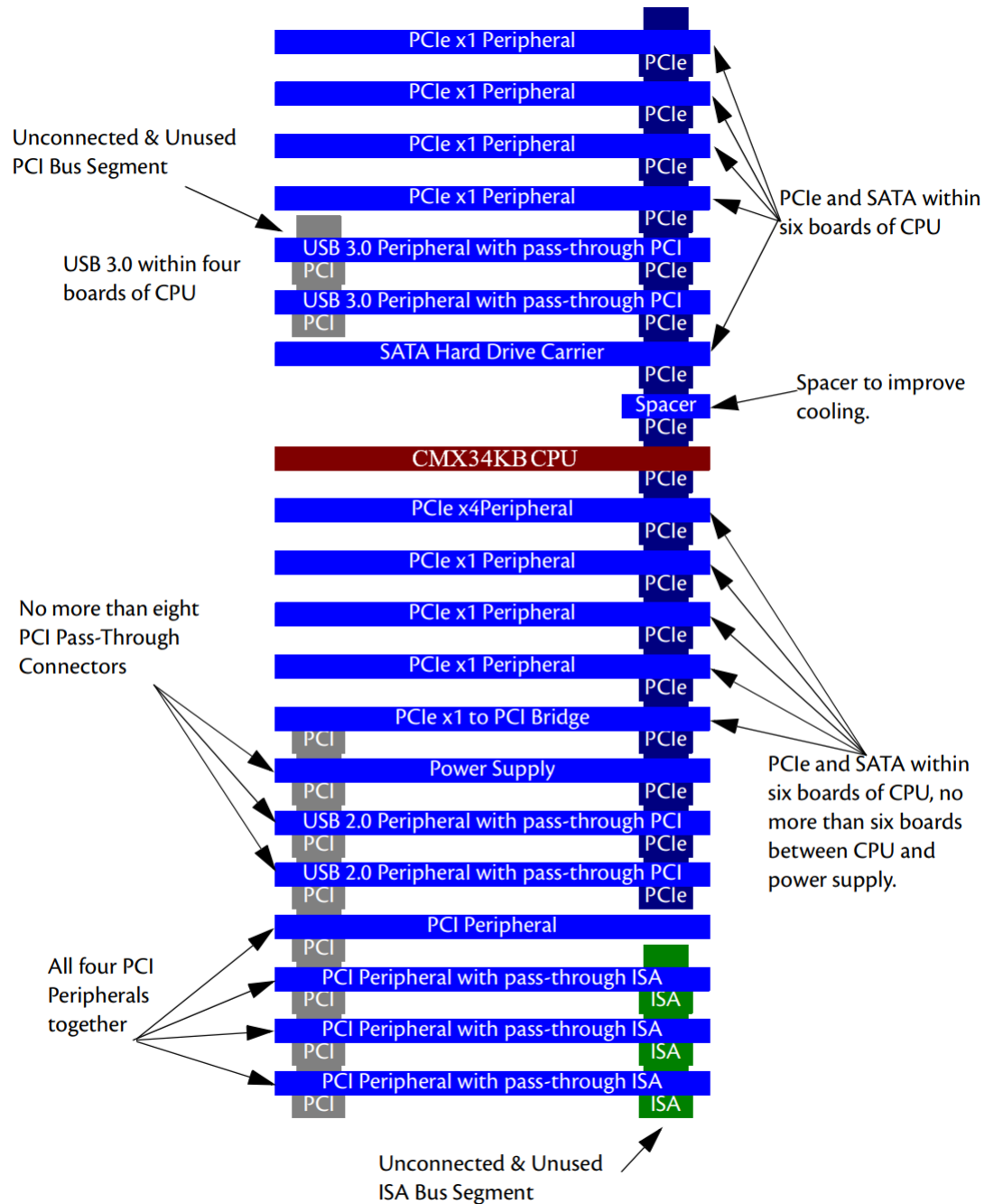


Figure 4 System Stacking Example

Connecting to the Stack

The bus connectors of the cpuModule are simply plugged onto a PC/104 stack to connect to other devices. Follow the procedure below to ensure that stacking of the modules does not damage connectors or electronics.



WARNING Do not force the module onto the stack! Wiggling the module or applying too much pressure may damage it. If the module does not readily press into place, remove it, check for bent pins or out-of-place keying pins, and try again.

For mechanical dimensions, including board-to-board spacing, see *Physical Dimensions*.

1. Turn off power to the PC/104 system or stack.
2. Always work at an ESD protected workstation and wear a grounded wrist-strap.
3. Select and install stand-offs to properly position the cpuModule on the stack.
4. Remove the cpuModule from its anti-static bag.
5. Check that pins of the bus connector are properly positioned.
6. Check the stacking order; make sure all the busses used by the peripheral cards are connected to the cpuModule.
7. Hold the cpuModule by its edges and orient it so the bus connector pins line up with the matching connector on the stack.
8. Gently and evenly press the cpuModule onto the PC/104 stack.

Power Input Connections

Power to the board must come from either the top or bottom PCIe/104 Type 2 bus connectors (**CN1** or **CN2**), or the auxiliary power connector (**CN3**). These connectors provide the required +12V DC voltage rail input to the cpuModule.

Connecting to the Utility Port 2.0 Connector

The Utility Port 2.0 connector (**CN5**) implements the following interfaces:

- One USB 2.0 (Universal Serial Bus) port
- Speaker port (0.1 W output)
- Hardware Reset input
- Battery input for Real Time Clock
- Soft Power Button input

To use these interfaces, you must connect to the Utility Port 2.0 connector (**CN5**). The Utility Port 2.0 cable from the RTD cable kit provides a small speaker, two USB 2.0 ports for a keyboard and mouse, a push-button for resetting the system, a soft-power button, and a lithium battery to provide backup power for the real time clock.

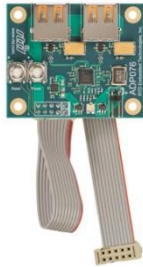


Figure 1 Utility Port 2.0 Cable

Refer to *Utility Port 2.0 Connector (CN5)* to connect devices to the Utility Port 2.0 connector.

Booting the CMX34KB/CMA34KB cpuModule for the First Time

You can now apply power to the cpuModule. If you press **ESC** on a keyboard while booting, the cpuModule will enter Setup. Once you have configured the cpuModule using Setup, save your changes and reboot. If you don't press **ESC**, the cpuModule will boot using the current settings.



Note You may miss the initial sign-on messages if your monitor takes a while to power on.

Note By default, cpuModules are shipped with a UEFI (Unified Extensible Firmware Interface) shell. When no other bootable device is installed in the system, the system will boot to it exclusively. This internal bootable shell may be disabled in the BIOS screen's boot menu.



WARNING Incorrect modifications to the BIOS can cause your system to malfunction. If you are unsure of any changes made to the BIOS use the **Load RTD Defaults** option under the **Exit** tab in the **SCU**. This option restores the BIOS to the default factory settings.

Chapter 3 Connecting the cpuModule

This chapter provides information on all CMX34KB/CMA34KB cpuModule connectors.

Proper Grounding Techniques – Page 35

Connector Locations – Page 35

Auxiliary Power (CN3) – Page 37

Utility Port 2.0 Connector (CN5) – Page 39

DisplayPort Connector (CN4) – Page 41

Serial Port 1 (CN7) and Serial Port 2 (CN8) – Page 43

Advanced Digital I/O (aDIO™) Port (CN6) – Page 49

Ethernet (10/100/1000Base-T and -TX) Connector (CN20, CN30) – Page 50

PCIe/104 Type 2 Bus (CN1 - Top and CN2 - Bottom) – Page 51

Optional RTC Battery Input (CN13) – Page 55

SATA 3.0 (CN15) – Page 55

USB 2.0 Connector (CN27) – Page 56

USB 3.0 Connector (CN11) – Page 56

Fan Connector (CN12) – Page 57

Proper Grounding Techniques

Before removing the CMX34KB/CMA34KB from its static bag, proper grounding techniques must be used to prevent electrostatic discharge (ESD) damage to the cpuModule. Common grounding procedures include an anti-static mat on a workbench, which may connect to an anti-static wrist strap (also known as an ESD wriststrap) on the wrist of the technician or engineer.

Connector Locations

Figure 5 shows the connectors of the CMX34KB cpuModule.

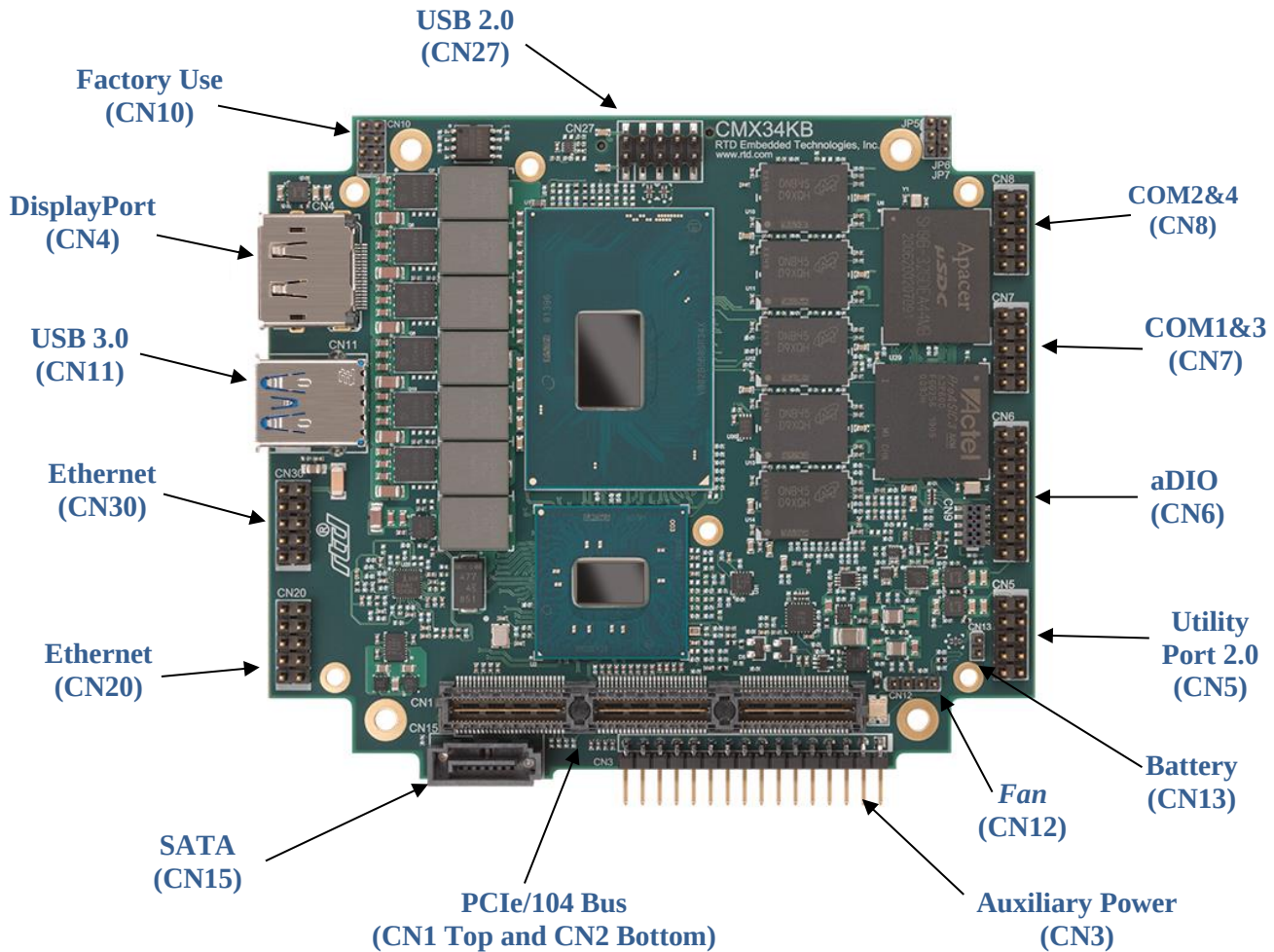


Figure 5 CMX34KB cpuModule (top view)



Note Pin 1 of each connector is indicated by a white silk-screened square on the top side of the board and a square solder pad on the bottom side of the board. Pin 1 of the bus connectors match when stacking PC/104 modules.

Figure 6 shows the connectors of the CMA34KB cpuModule.

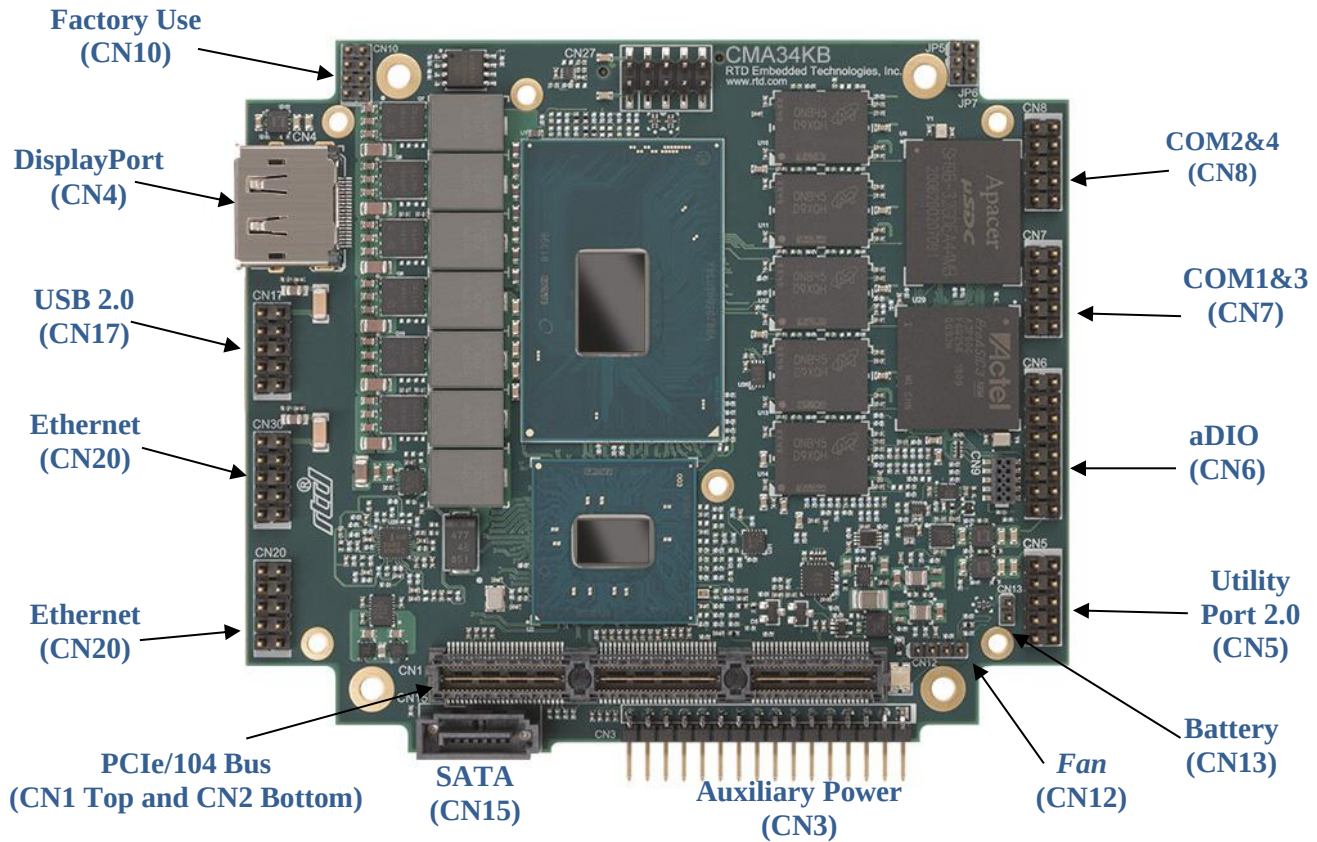


Figure 6 CMA34KB cpuModule (top view)

Table 6 CMX34KB/CMA34KB Basic Connectors and Mating Connectors

Connector	Function	Size and Pitch	Mating Connector
CN1	PCIe/104 Type 2 Bus (Top)	156-pin, 0.635mm	Samtec ASP-129646-03
CN2	PCIe/104 Type 2 Bus (Bottom)	156-pin, 0.635mm	Samtec ASP-129637-03
CN3	Auxiliary Power	1x16, 0.1"	FCI 65039-021LF (housing model number only)
CN4	DisplayPort	Molex Defined	Molex P/N Series 47272-0001
CN5	Utility Port 2.0	2x5, 0.1"	3M 89110-0001
CN6	aDIO	2x8, 0.1"	3M 89116-0001
CN7	Serial Port 1 (COM1&3)	2x5, 0.1"	3M 89110-0001
CN8	Serial Port 2 (COM2&4)	2x5, 0.1"	3M 89110-0001
CN11 ¹	USB 3.0	Samtec USB Series USB3-A	Samtec USB3-A-S-S-TH
CN12	Fan (reserved)	1x4, 2mm	FCI 69305-004LF
CN13	Battery Connector	1x2, 2mm	TMM-102-03-G-S

CN15	SATA 3.0	Molex defined	Molex P/N 678005005
CN17²	USB 2.0	2x5, 0.1"	3M 89116-0001
CN27	USB 2.0	2x5, 0.1"	3M 89110-0001
CN20 and CN30	Ethernet	2x5, 0.1"	3M 89110-0001

1. Only on CMX34KB. Replaces CN17.
2. Only on CMA34KB. Replaces CN11.

Auxiliary Power (CN3)

The Auxiliary Power connector (**CN3**) can be used to supply power to devices that are attached to the cpuModule. These devices include hard drive, front-end boards for data acquisition systems, and other devices.

Power can also be conveyed to the module through the Auxiliary Power connector (**CN3**). The Intel Xeon Series cpuModule requires +12 VDC and ground for operation.



Note Although it is possible to power the cpuModule through the Auxiliary Power connector, the preferred method is to power it through the bus connector from a power supply in the stack. The cpuModule can have large current transients during operation, which make powering it through wires difficult. Powering through the bus eliminates problems such as resistive voltage drop and lead inductance transients.



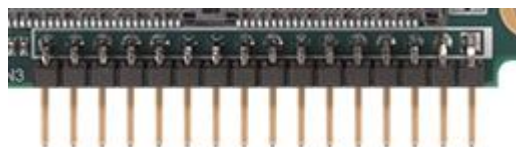
WARNING There is no reverse voltage protection on CN3. Reverse connecting CN3 will void the warranty. Always check polarity before applying power.

If using the Auxiliary Power connector to power the system, care must be taken to ensure good power connections. The power and ground leads must be twisted together, or as close together as possible to reduce lead inductance. A separate lead must be used for each of the power pins. All 5V pins and all ground pins must be connected. Do not use wire smaller than 20 gauge, and take care to ensure the length of the wire does not exceed 2 ft. The power supply solution must be verified by measuring voltage at the Auxiliary Power Connector and verifying that it meets the input voltage specifications. The voltage at the connector should be checked with an oscilloscope while the system is operational.



WARNING This 12-pin power connector is **not** compatible with some older generations of RTD cpuModules.

WARNING If you connect power incorrectly, the module will almost certainly be destroyed. Please verify power connections to the module before applying power.



(CN3) Pin 1

Table 7 Auxiliary Power Connector (CN3)

Pin	Signal	Function
1	GND	Ground
2	+12 V	+12 Volts DC
3	GND	Ground
4	+5 V	+5 Volts DC
5	+5 V	+5 Volts DC
6	GND	Ground
7	GND	Ground
8	+12 V ¹	+12 Volts DC
9	+12 V ¹	+12 Volts DC
10	GND	Ground
11	GND	Ground
12	+5 V	+5 Volts DC
13	+5 V	+5 Volts DC
14	GND	Ground
15	+12 V	+12 Volts DC
16	GND	Ground



WARNING The user is responsible for any over-voltage and reverse voltage protection that would be necessary to power through CN3. Over-voltage damage at CN3 is not covered under warranty.

Utility Port 2.0 Connector (CN5)

The Utility Port 2.0 connector implements the following functions:

- 1x USB 2.0 (Universal Serial Bus) Port
- Speaker port (0.1 W output)
- Hardware Reset input
- Soft Power Button input
- Battery input for Real Time Clock

Table 8 provides the pinout of the Utility Port 2.0 connector.

Table 8 Utility Port 2.0 Connector (CN5)

Pin	Signal	Function	In/Out
1	SPKR	Speaker Output (open collector)	out
2	PWR	+5 V	out
3	RESET#	Manual Push-Button Reset	in
4	DATA-	Bidirectional data line for USB1	in/out
5	PWRSW#	Soft Power Button	in
6	DATA+	Bidirectional data line for USB1	in/out
7	GND	Ground	out
8	GND	Ground	out
9	BAT	RTC Battery Input	in
10	Shield GND	Shield Ground	out



WARNING The pinout of the Utility Port 2.0 connector is **not** compatible with some older generations of RTD cpuModules. Attaching a legacy Utility Port harness to the Utility Port 2.0 connector may damage or destroy the cpuModule.

Facing into the connector pins, the pinout is shown in the table below: (pin one is indicated by a square solder pad and white silk screen)

9	7	5	3	1
BAT	GND	PWRSW#	RESET#	SPKR
Shield GND	GND	DATA+	DATA-	PWR
10	8	6	4	2



(CN5) Pin 1

USB 2.0 Connector

One USB 2.0 compliant connector is available on connector **CN5**. Table 8 provides the pinout of the USB connector.

Speaker

A speaker output is available on pins 1 and 2 of the Utility Port 2.0 connector. These outputs are controlled by a transistor to supply 0.1 W of power to an external speaker. The external speaker should have 8 Ohm impedance and be connected between pins 1 and 2.

System Reset

Pin 3 of the Utility Port 2.0 connector allows connection of an external push-button to manually reset the system. The push-button is normally open and connected to ground when pushed. The type of reset generated by this button can be set in the BIOS configuration utility.

Soft Power Button

Pin 5 of the Utility Port 2.0 connector allows connection of an external push-button to send a soft power signal to the system. The push-button is normally open and connected to ground when pushed. For more information on the modes of the Soft Power Button, refer to the *Power Management* section in Chapter 4, *Using the cpuModule*

Battery

Pin 9 of the Utility Port 2.0 connector is the connection for an external backup battery. The cpuModule uses this battery to preserve the date and time of the real time clock when system power is removed.

Connecting a battery is only required to maintain time when power is completely removed from the cpuModule. A battery is not required for board operation.



WARNING The optional RTC battery input connector (**CN13**) should be left unconnected if the multi-function connector (**CN5**) has a battery connected to pin 9.

DisplayPort Connector (CN4)

The DisplayPort connector on the CMX34KB/CMA34KB cpuModule is a standard PC DisplayPort connector complete with latch holes to provide a rugged connecting solution for latching DisplayPort cables.

The DisplayPort supports all mandatory features of the VESA DisplayPort (DP) 1.4 standard and supports audio over the connection. Full bandwidth transmission is supported over a 2-meter cable, with reduced bandwidth transmission supported up to a length of 15 meters.

The connection can directly output single-link HDMI and DVI signals by using a simple passive adapter. Passive adapters achieve HDMI and DVI signaling by adjusting the lower signal levels output by the connector.

Table 9 provides the pinout of the DisplayPort connector.

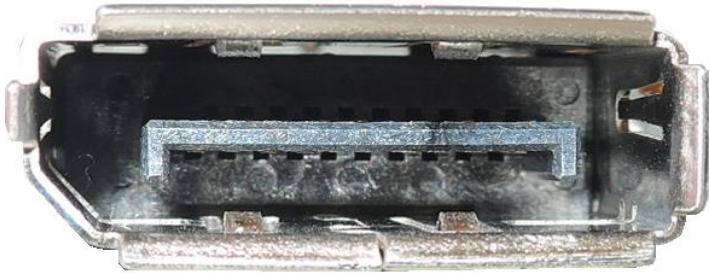
Table 9 DisplayPort Connector (CN4)

Pin	Signal	Standard Function	DVI / HDMI mode ¹	In/Out
1	LN0+	Main Link, Lane 0 (positive)	TMDS Channel 2 (positive)	out
2	GND	Ground	Ground	out
3	LN0-	Main Link, Lane 0 (negative)	TMDS Channel 2 (negative)	out
4	LN1+	Main Link, Lane 1 (positive)	TMDS Channel 1 (positive)	out
5	GND	Ground	Ground	out
6	LN1-	Main Link, Lane 1 (negative)	TMDS Channel 1 (negative)	out
7	LN2+	Main Link, Lane 2 (positive)	TMDS Channel 0 (positive)	out
8	GND	Ground	Ground	out
9	LN2-	Main Link, Lane 2 (negative)	TMDS Channel 0 (negative)	out
10	LN3+	Main Link, Lane 3 (positive)	TMDS Clock (positive)	out
11	GND	Ground	Ground	out
12	LN3-	Main Link, Lane 3 (negative)	TMDS Clock (negative)	out
13	CFG1	Configuration Pin 1	Cable Adapter Detect	out
14	CFG2	Configuration Pin 2	Consumer Electronics Control ²	out
15	AUX+	Auxiliary Channel (positive)	DDC Clock	in/out
16	GND	Ground	Ground	out
17	AUX-	Auxiliary Channel (negative)	DDC Data	in/out
18	HPD	Hot Plug Detect	Hot Plug Detect	in
19	DPG	Return for DPV	Return for DPV	out
20	DPV	+3.3V DC Power	+3.3V DC Power	out

1. Requires special passive adapter

2. HDMI mode only

Facing the connector pins of the DisplayPort connector (CN4), the pinout is:



19	17	15	13	11	9	7	5	3	1
DPG	AUX-	AUX+	CFG1	GND	LN2-	LN2+	GND	LN0-	LN0+
DPV	HPD	GND	CFG2	LN3-	LN3+	GND	LN1-	LN1+	GND
20	18	16	14	12	10	8	6	4	2

Serial Port 1 (CN7) and Serial Port 2 (CN8)

Serial Port 1 (COM1) is implemented on connector **CN7**, and Serial Port 2 (COM2) is implemented on connector **CN8**. The serial ports are normally configured as PC compatible full-duplex RS-232 ports, but you may use the BIOS Setup program to reconfigure these ports as half-duplex RS-422 or full-duplex RS-422 or RS-485. If you reconfigure the ports, you must also select the I/O address and corresponding interrupt using Setup. Table 10 provides the standard I/O addresses and corresponding interrupts.

Table 10 Serial Port Settings

I/O Address (hex)	IRQ
03F8	IRQ4
02F8	IRQ3
03E8	IRQ4
02E8	IRQ3

Serial Port UART

The serial ports are implemented with a 16550/16750-compatible UART (Universal Asynchronous Receiver/Transmitter). This UART is capable of baud rates up to 115.2 KBaud and includes a 512-byte FIFO. Refer to any standard PC-AT hardware reference for the register map of the UART.

It is possible to change the slew rate for the UARTs of the cpuModule to allow the serial ports to operate at higher speeds than 115200 bps. For more information see [Chapter 4 - “Non-Standard Serial Port Modes”](#).

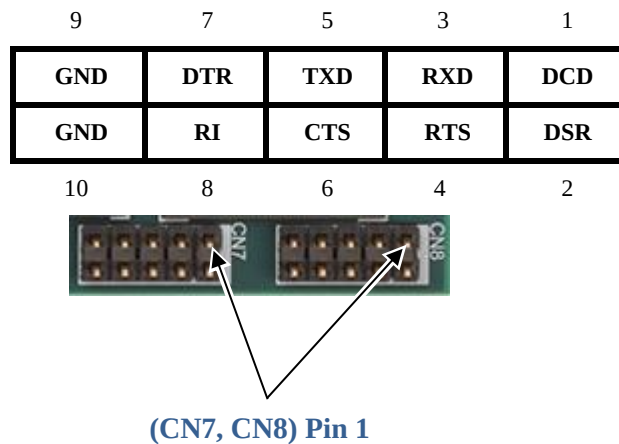
RS-232 Serial Port (Default)

The default serial port mode is full-duplex RS-232. With this mode enabled, the serial port connectors must be connected to RS-232 compatible devices. Table 11 provides the serial port connector pinout and shows how to connect to an external DB-25 or DB-9 compatible serial connector.

Table 11 Serial Port in RS-232 Mode

Pin	Signal	Function	In/Out	DB-25	DB-9
1	DCD	Data Carrier Detect	in	8	1
2	DSR	Data Set Ready	in	6	6
3	RXD	Receive Data	in	3	2
4	RTS	Request To Send	out	4	7
5	TXD	Transmit Data	out	2	3
6	CTS	Clear To Send	in	5	8
7	DTR	Data Terminal Ready	out	20	4
8	RI	Ring Indicate	in	22	9
9	GND	Signal Ground	—	7	5
10	GND	Signal Ground	—	—	—

Facing the serial port's connector pins, the pinout is:



RS-422 or RS-485 Serial Port

You may use the BIOS setup utility to configure the serial ports as RS-422 or RS-485. In this case, you must connect the serial port to an RS-422 or RS-485 compatible device.

When using RS-422 mode, you can use the serial ports in either half-duplex (two-wire) or full-duplex (four-wire) configurations.



Note The cpuModule has a 120 Ohm termination resistor. Termination is usually necessary on all RS-422 receivers and at the ends of the RS-485 bus. Termination resistors can be enabled in the BIOS setup utility.

When using full-duplex in RS-422 mode, connect the ports as shown in Table 12.

Table 12 Full-Duplex Connections

Port 1	Port 2
RXD+	TXD+
TXD+	RXD+
RXD-	TXD-
TXD-	RXD-

In RS-485 mode, the connection of the ports is always half-duplex, as the transceivers' transmitters are connected to the receivers internally.

RS-422 Mode Pinout

Table 13 provides the serial port connector pinout when RS-422 mode is enabled.

Table 13 Serial Port in RS-422 Mode

Pin	Signal	Function	In/Out	DB-9
1	TXD-	Transmit Data (-)	out	1
2	—	Reserved	—	6
3	TXD+	Transmit Data (+)	out	2
4	—	Reserved	—	7
5	RXD+	Receive Data (+)	in	3
6	—	Reserved	—	8
7	RXD-	Receive Data (-)	in	4
8	—	Reserved	—	9
9	GND	Signal Ground	—	5
10	GND	Signal Ground	—	—

Facing the serial port connector, the pinout is:

9	7	5	3	1
GND	RXD-	RXD+	TXD+	TXD-
GND	Rsvd	Rsvd	Rsvd	Rsvd
10	8	6	4	2



WARNING The pinout of the COM ports in RS-422 mode is **not** compatible with some older generations of RTD cpuModules.

RS-485 Mode Pinout

Table 14 provides the serial port connector pinout when RS-485 mode is enabled

Table 14 Serial Port in RS-485 Mode

Pin	Signal	Function	In/Out	DB-9
1	D-	Data (-)	in/out	1
2	—	Reserved	—	6
3	D+	Data (+)	in/out	2
4	—	Reserved	—	7
5	—	Reserved	—	3
6	—	Reserved	—	8
7	—	Reserved	—	4
8	—	Reserved	—	9
9	GND	Signal Ground	—	5
10	GND	Signal Ground	—	—

Facing the serial port connector, the pinout is:

9	7	5	3	1
GND	Rsvd	Rsvd	D+	D-
GND	Rsvd	Rsvd	Rsvd	Rsvd
10	8	6	4	2



WARNING The pinout of the COM ports in RS-485 mode is **not** compatible with some older generations of RTD cpuModules.



Note When using the serial port in RS-485 mode, the serial transmitters are enabled and disabled under software control. The transmitters can be enabled by manipulating the RequestTo Send (RTS*) signal of the serial port controller. This signal is controlled by writing bit 1 of the Modem Control Register (MCR) as follows:

- If MCR bit 1 = 1, then RTS* = 0, and serial transmitters are disabled
- If MCR bit 1 = 0, then RTS* = 1, and serial transmitters are enabled

Note For more information on the serial port registers, including the MCR, refer to the Serial Port Programming reference in Appendix D.

Dual Serial Port Modes

The serial port connectors can be configured as dual serial ports in the BIOS. This enhancement in dual RS-232 comes at the cost of losing the DSR and DTR signals, which are only available in single RS-232 mode.

The mapping between the connectors and COM port numbers is shown in Table 15. The supported combinations of serial port modes are listed in Table 16, which also includes a reference to the corresponding connector pinout.

Table 15 Dual Serial Port Connections

Connector	COM A	COM B
CN7	COM 1	COM 3
CN8	COM 2	COM 4

Table 16 Dual Serial Port Modes

COM A	COM B	Pinout Reference	Echo Mode
RS-232	RS-232	Table 18	—
RS-422	RS-422	Table 19	—
RS-485	RS-485	Table 20	echo canceled
RS-485	RS-485	Table 19	with echo ¹

1. Dual RS-485 mode with echo uses dual RS-422 mode pinout



WARNING The dual serial port mode pinouts of the COM ports are **not** compatible with some older generations of RTD cpuModules.



Note The cpuModule has 120 Ohm termination resistors. Termination is usually necessary on all RS-422 receivers and at the ends of the RS-485 bus. Termination resistors can be enabled in the BIOS setup utility. When termination is enabled in dual port mode, it is enabled for both ports.

Table 17 COM A (RS-232) and COM B (RS-232)

Pin	Signal	Function	In/Out	DB-9
1	DCD1	COM A - Data Carrier Detect	in	1
2	CTS1	COM A - Clear To Send	in	6
3	RXD1	COM A - Receive Data	in	2
4	TXD2	COM B - Transmit Data	out	7
5	TXD1	COM A - Transmit Data	out	3
6	RXD2	COM B - Receive Data	in	8
7	RTS1	COM A - Request To Send	out	4

Table 18 COM A (RS-232) and COM B (RS-232)

Pin	Signal	Function	In/Out	DB-9
8	RI1	COM A - Ring Indicate	in	9
9	GND	Signal Ground	—	5
10	GND	Signal Ground	—	—

Table 19 COM A (RS-422) and COM B (RS-422)¹

Pin	Signal	Function	In/Out	DB-9
1	TXD1-	COM A - Transmit Data (–)	out	1
2	TXD2-	COM B - Transmit Data (–)	out	6
3	TXD1+	COM A - Transmit Data (+)	out	2
4	TXD2+	COM B - Transmit Data (+)	out	7
5	RXD1+	COM A - Receive Data (+)	in	3
6	RXD2+	COM B - Receive Data (+)	in	8
7	RXD1-	COM A - Receive Data (–)	in	4
8	RXD2-	COM B - Receive Data (–)	in	9
9	GND	Signal Ground	—	5
10	GND	Signal Ground	—	—

1. Dual RS-485 mode with echo uses dual RS-422 mode pinout

Table 20 COM A (RS-485) and COM B (RS-485)

Pin	Signal	Function	In/Out	DB-9
1	D1-	COM A - Data (–)	in/out	1
2	D2-	COM B - Data (–)	in/out	6
3	D1+	COM A - Data (+)	in/out	2
4	D2+	COM B - Data (+)	in/out	7
5	—	Reserved	—	3
6	—	Reserved	—	8
7	—	Reserved	—	4
8	—	Reserved	—	9
9	GND	Signal Ground	—	5
10	GND	Signal Ground	—	—

Advanced Digital I/O (aDIO™) Port (CN6)

Connector **CN6** is configured as an aDIO port. aDIO is 12 digital bits configured as 8-bit programmable and 4-bit port programmable I/O, providing any combination of inputs and outputs. Match, event, and strobe interrupt modes mean no more wasting valuable processor time polling digital inputs. Interrupts are generated when the 8-bit programmable digital inputs match a pattern, or on any value change event. Bit masking allows selecting any subgroup of 8 bits. The strobe input latches data into the bit programmable port and generates an interrupt. Refer to *Advanced Digital I/O Ports (aDIO™)* —page 64 for information on programming the aDIO.

Table 21 aDIO Pinout

CN6 Pin	Function	CN6 Pin	Function
1	P0-0	2	P0-1
3	P0-2	4	P0-3
5	P0-4	6	P0-5
7	P0-6	8	P0-7
9	strobe 0	10	strobe 1
11	P1-0	12	P1-1
13	P1-2	14	P1-3
15	GND	16	+5 V ¹

1. Available during standby.



(CN6) Pin 1

Ethernet (10/100/1000Base-T and -TX) Connector (CN20, CN30)

This connector provides a 10/100/1000Base-T Ethernet connection. Table 22 provides the pinout of the Ethernet connector. For 1000Base-T, all four pairs are used for transmit and receive.

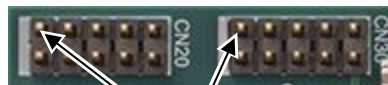
To use the onboard 10/100/1000 Ethernet controller, Ethernet must be enabled in the BIOS.

When enabled, the multi-color LED will blink to indicate an Ethernet connection. For more information, refer to the *Multi-Color LED* section.

Table 22 Ethernet Connector (CN20 and CN30)

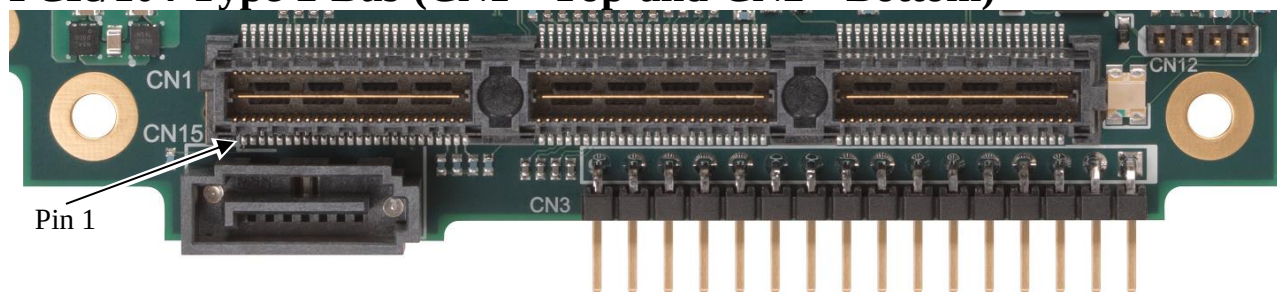
RJ-45 Pin	10-Pin DIL Pin	Signal	Function
3	1	B+ (RX+)	Receive+ (10/100)
6	2	B- (RX-)	Receive- (10/100)
4	3	C+	
5	4	C-	
1	5	A+ (TX+)	Transmit+ (10/100)
2	6	A- (TX-)	Transmit- (10/100)
7	7	D+	
8	8	D-	
—	9	AGND	Ethernet Ground
—	10	AGND	Ethernet Ground

9	7	5	3	1
AGND	D+	A+	C+	B+
AGND	D-	A-	C-	B-
10	8	6	4	2



(CN20, CN30) Pin 1

PCIe/104 Type 2 Bus (CN1 - Top and CN2 - Bottom)



Connectors **CN1** and **CN2** carry the signals of the PCIe/104 PCIe bus. These signals match definitions found in the **PCI/104-Express & PCIe/104 Specification Version 3.1** from the PC/104 Embedded Consortium. Table 23 lists the pinouts of the PC/104-Express bus connector.



WARNING Not all PCIe cards are compatible with the PCIe/104 Type 2 connector. Be sure that all boards attached to this bus are compatible before powering the system.

Table 23 PCIe/104 Type 2 Bus Signal Assignments ¹

Pin	Signal	Signal	Pin
1	USB_OC#	PE_RST#	2
3	+3.3V ²	+3.3V ²	4
5	USB_1p	USB_0p	6
7	USB_1n	USB_0n	8
9	GND	GND	10
11	PEx1_1Tp	PEx1_0Tp	12
13	PEx1_1Tn	PEx1_0Tn	14
15	GND	GND	16
17	PEx1_2Tp	PEx1_3Tp	18
19	PEx1_2Tn	PEx1_3Tn	20
21	GND	GND	22
23	PEx1_1Rp	PEx1_0Rp	24
25	PEx1_1Rn	PEx1_0Rn	26
27	GND	GND	28
29	PEx1_2Rp	PEx1_3Rp	30
31	PEx1_2Rn	PEx1_3Rn	32
33	GND	GND	34
35	PEx1_1Clkp	PEx1_0Clkp	36
37	PEx1_1Clkn	PEx1_0Clkn	38
39	+5V_STBY	+5V_STBY	40
41	PEx1_2Clkp	PEx1_3Clkp	42
43	PEx1_2Clkn	PEx1_3Clkn	44
45	CPU_DIR	PWRGOOD	46

+5 Volts

47	Reserved	Reserved	48
49	Reserved	Reserved	50
51	Reserved	PSON#	52

Table 24 **PCIe/104 Type 2 Bus Signal Assignments ¹**

Pin	Signal	Signal	Pin
53	WAKE#	STK1	54
55	GND	GND	56
57	PEx4_1T(0)P	PEx4_0T(0)P	58
59	PEx4_1T(0)N	PEx4_0T(0)N	60
61	GND	GND	62
63	PEx4_1T(1)P	PEx4_0T(1)P	64
65	PEx4_1T(1)N	PEx4_0T(1)N	66
67	GND	GND	68
69	PEx4_1T(2)P	PEx4_0T(2)P	70
71	PEx4_1T(2)N	PEx4_0T(2)N	72
73	GND	GND	74
75	PEx4_1T(3)P	PEx4_0T(3)P	76
77	PEx4_1T(3)N	PEx4_0T(3)N	78
79	GND	GND	80
81	SATA_T1_P	SATA_T0_P	82
83	SATA_T1_N	SATA_T0_N	84
85	GND	GND	86
87	USB3_T1_P	USB3_T0_P	88
89	USB3_T1_N	USB3_T0_N	90
91	GND	GND	92
93	Reserved	Reserved	94
95	Reserved	Reserved	96
97	GND	GND	98
99	Reserved	Reserved	100
101	Reserved	Reserved	102
103	GND	GND	104

+5 Volts

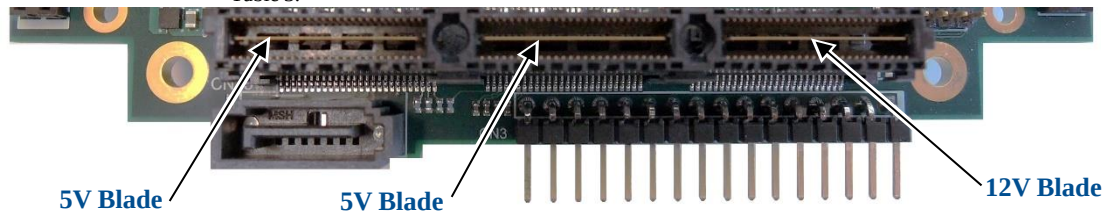
Table 24 PCIe/104 Type 2 Bus Signal Assignments ¹

Pin	Signal	Signal	Pin
105	STK2	Reserved	106
107	GND	GND	108
109	PEx4_1R(0)p	PEx4_0R(0)p	110
111	PEx4_1R(0)n	PEx4_0R(0)n	112
113	GND	GND	114
115	PEx4_1R(1)p	PEx4_0R(1)p	116
117	PEx4_1R(1)n	PEx4_0R(1)n	118
119	GND	GND	120
121	PEx4_1R(2)p	PEx4_0R(2)p	122
123	PEx4_1R(2)n	PEx4_0R(2)n	124
125	GND	GND	126
127	PEx4_1R(3)p	PEx4_0R(3)p	128
129	PEx4_1R(3)n	PEx4_0R(3)n	130
131	GND	GND	132
133	SATA_1Rp	SATA_0Rp	134
135	SATA_1Rn	SATA_0Rn	136
137	GND	GND	138
139	USB3_R1_P	USB3_R0_P	140
141	USB3_R1_N	USB3_R0_N	142
143	GND	GND	144
145	Reserved	Reserved	146
147	Reserved	Reserved	148
149	GND	GND	150
151	Reserved	Reserved	152
153	Reserved	Reserved	154
155	GND	GND	156

+12 Volts

1. Signals marked with (#) are active low.

2. +3.3V is not required for board operation. For power input requirements, see Table 3.

**Figure 6 Locations of 5V and 12V Power Blades**

PCIe/104 Type 2 Compatibility

The PCIe/104 Type 2 connector is compatible with any PCI/104-Express or PCIe/104 peripheral module that does not use the x16 Link. This includes any card that uses the PCIe x1 links, USB, or a power supply. In addition, this connector can be used to add SATA devices to the system.

If a card is installed that is not compatible with the Type 2 connector, the CPU will keep the system in soft-off, and the LED will be Cyan to indicate that there is a Bus Stacking Error.

PCIe Link Configuration

This cpuModule supports a total of eight PCIe x1 links and four PCIe x4 links for system expansion. The PCIe links are split between the top-side PCIe/104 connector (CN1) and bottom-side PCIe/104 connector (CN2) as seen in Table 25.

Table 25 PCIe Link Configuration

Link Width	Connector	Number of Links per Connector
x1	CN1, CN2	4
x4	CN1, CN2	2

Optional RTC Battery Input (CN13)

The optional RTC battery input is the connection for an external backup battery. This battery is used by the cpuModule when system power is removed to preserve the date and time of the real time clock.

Connecting a battery is only required to maintain time when power is completely removed from the cpuModule. A battery is not required for board operation.

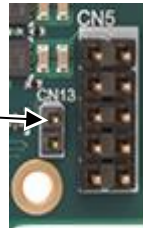
Table 26 **Optional RTC Battery Input (CN13)**

Pin	Signal	Function
1	BAT	RTC Battery Input
2	GND	Ground



WARNING This optional RTC battery connector (**CN13**) should be left unconnected if the Utility Port 2.0 connector (**CN5**) has a battery connected.

(CN13) Pin 1



SATA 3.0 (CN15)

CN15 is a SATA 3.0 connection direct to the PCH.



USB 2.0 Connector (CN27)

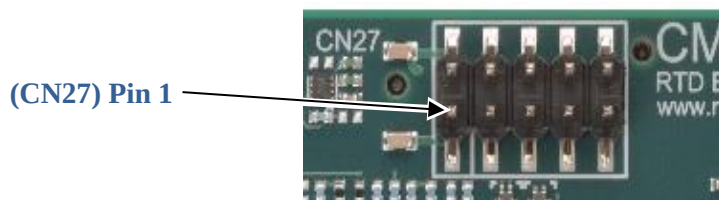
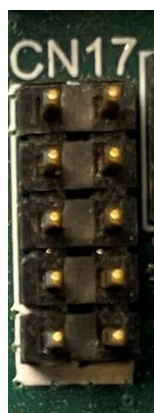


Table 27 USB Connector (CN17 and CN27)

Pin	Signal	Function	In/Out
1	VCC1	Supply +5V to USB1	out
2	VCC2	Supply +5V to USB1	out
3	DATA1-	Bidirectional data line for USB1	in/out
4	DATA2-	Bidirectional data line for USB2	in/out
5	DATA1+	Bidirectional data line for USB1	in/out
6	DATA2+	Bidirectional data line for USB2	in/out
7	GND	Ground	out
8	GND	Ground	out
9	Shield	Shield Ground	—
10	Shield	Shield Ground	—

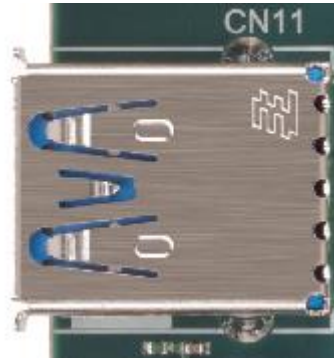
USB 2.0 Connector (CN17)

Populated on the **CMA34KB** only. 10-pin DIL is USB 2.0. See Table 27 for the pinout.



USB 3.0 Connector (CN11)

Populated on the **CMX34KB** only. The Type A receptacle is USB 3.0.



Fan Connector (CN12)

For factory use. Do not connect anything but the RTD supplied fan for the cpuModule.

Chapter 4 Using the cpuModule

This chapter provides information for users who wish to develop their own applications programs for the CMX34KB/CMA34KBs cpuModule. This chapter includes information on the following topics:

The RTD Enhanced BIOS – Page 59

Operating System Specific Usage – Page 62

Non-Standard Serial Port Modes – Page 63

Advanced Digital I/O Ports (aDIO™) – Page 64

SATA Controller Configuration – Page 68

Watchdog Timer Control – Page 69

Thermal Management – Page 70

Power Management – Page 71

Multi-Color LED – Page 73

Reset and Event Status Register – Page 74

Trusted Platform Module – Page 76

Features and Settings That Can Affect Boot Time – Page 77

System Recovery – Page 78

The RTD Enhanced BIOS

The RTD Enhanced BIOS is software that interfaces hardware-specific features of the cpuModule to an operating system (OS). Physically, the BIOS software is stored in a Flash EPROM on the cpuModule. Functions of the BIOS are divided into two parts.

The first part of the BIOS is known as POST (power-on self-test) software, and it is active from the time power is applied until an OS boots (begins execution). POST software performs a series of hardware tests, sets up the machine as defined in Setup, and begins the boot of the OS.

The second part of the BIOS is known as the CORE BIOS. It is the normal interface between cpuModule hardware and the OS which is in control. It is active from the time the OS boots until the cpuModule is turned off. The CORE BIOS provides the system with a series of software interrupts to control various hardware devices.

Configuring the RTD Enhanced BIOS

The cpuModule Setup program allows you to customize the cpuModule's configuration. Selections made in Setup are stored on the board and are read by the BIOS at power-on.

Entering the BIOS Setup through the Graphical BIOS Menu

You can run Setup by rebooting the cpuModule and repeatedly pressing the **ESC** key. This will bring to you to a graphical BIOS menu with several options described in the table below.

To enter Setup, navigate to the **Setup Utility** option with the arrow keys, and press **Enter**.



Table 28 Graphical BIOS Menu Options

Key	Function
Continue	Exit the graphical menu, and proceed with booting
Boot Manager	Select a device to boot to during the current power cycle, and override the selected boot device order in the Setup
Device Management	Set the primary video BIOS during the current power cycle, and view a list of the system's detected devices (disk, video, network, input, etc.)
Boot From File	Displays a list of bootable files in the system
Administer Secure Boot	Traditional BIOS Setup

Table 29 Graphical BIOS Menu Options

Key	Function
SCU	Traditional BIOS Setup

Field Selection

To move between fields in Setup, use the keys listed below. When you are finished with Setup, save your changes and exit. The system will automatically reboot.

Table 30 Setup Keys

Key	Function
F1	Help
→, ←, ↓, ↑	Move between fields
F5, F6	Selects next/previous values in fields
Enter	Go to the submenu for the field
ESC	To previous menu then to exit menu
F8	Load Setup defaults
F10	Save settings and exit Setup

Main Menu Setup Fields

The following is a list of Main Menu Setup fields.

Table 31 Main Menu Setup Fields

Field	Active Keys	Selections
Main	Press Enter to select	Access system information such as the cpuModule's serial number, CPU speed and type, BIOS version, FPGA version, and CMOS time and date settings
Advanced	Press Enter to select	Setup advanced RTD cpuModule features, including boot options, aDIO and serial port configuration, and miscellaneous feature control.
Security	Press Enter to select	Setup the supervisor and access password
Boot	Press Enter to select	Set the system boot sequence
Power	Press Enter to select	Set up various power savings modes
Exit	Press Enter to select	Save or discard changes and exit the BIOS, or load the default BIOS settings



Note: Future BIOS versions may have slightly different setup menus and options.

Note: The **Main** page shows the cpuModule's **serial number**, **BIOS Version**, and **FPGA Version**. These numbers can be useful if you need techsupport for your cpuModule.



WARNING Incorrect modifications to the BIOS can cause your system to break. If you are unsure of any changes made to the BIOS, there is a **Load RTD Defaults** option under the **Exit** tab in the **SCU**. This option restores the BIOS to the default factory settings.

Operating System Specific Usage

Windows®

The CMX34KB/CMA34KB is fully supported under Windows 10. While Windows 7 isn't officially supported, it can work using a work-around. Both 32-bit and 64-bit editions are supported. Drivers for onboard peripherals such as video and Ethernet are provided on the companion DVD that is shipped with the board. Updated drivers may be available for download from the RTD website. (www.rtd.com)

Linux®

The CMX34KB/CMA34KB is fully supported with Linux kernel version 4.10 and onward. Earlier versions of the kernel may not contain drivers for all chipset features (most notably SATA), and may not work as expected.

At the time of publication, RTD has validated Ubuntu 20.04, OpenSuse 13.2, and OpenSuse LEAP 15.0. However, any standard PC Linux distribution with a 4.10 or newer kernel is expected to work.

UEFI OS Installations

When dealing with UEFI OS installations, the BIOS setting for “EFI/Legacy Device Order” will need to be changed to “EFI devices first”. The default boot order is to boot to “Legacy devices first”, which will render the OS unbootable after an installation.

Non-Standard Serial Port Modes

It is possible to change the input clock rate for the UARTs of the cpuModule to allow the serial ports to operate at higher speeds than 115,200 bps. This is accomplished by modifying the baud rate multiplier for each serial port.

Non-standard baud rates are supported for all COM port modes of the CMX34KB:

Table 32 Baud Rate Multiplier Registers

Connector	Port	Address Range (hex)
CN7 (A)	COM 1	EE0–EE1h
CN7 (B)	COM 3	EE2–EE3h
CN8 (A)	COM 2	EE4–EE5h
CN8 (B)	COM 4	EE6–EE7h

To set an alternate maximum baud rate for a serial port, write the value which corresponds to the desired multiplier to the port's baud rate multiplier register:

Table 33 Baud Rate Multipliers

Value	Maximum Baud Rate	
	bps	Multiplier
0x0E28	115,200	1x
0x38A0	460,800	4x
0x713F	921,600	8x
0x7AE1	1,000,000	8.7x



Note When using the non-standard high speed serial port modes, it is highly recommended to use hardware flow control, whenever possible.

Advanced Digital I/O Ports (aDIO™)

The aDIO has a maximum read/write operating speed of 16.66 KHz. This board supports 12 bits of TTL/CMOS compatible digital I/O (TTL signaling). These I/O lines are grouped into two ports, Port 0 and Port 1. Port 0 is bit programmable; Port 1 is byte programmable. Port 0 supports RTD's Advanced Digital Interrupt modes. The three modes are strobe, match, and event. Strobe mode generates an interrupt and latches Port 0 when the strobe input transitions from low to high. Match mode generates an interrupt when an 8-bit pattern is received in parallel that matches the match mask register. Event mode generates an interrupt when a change occurs on any bit. In any mode, masking can be used to monitor selected lines.

When the CPU boots, all digital I/O lines are programmed as inputs, meaning that the digital I/O line's initial state is undetermined. If the digital I/O lines must power up to a known state, an external 10 k-Ohm resistor must be added to pull the line high or low.

The 8-bit control read/write registers for the digital I/O lines are located from I/O address EC0h to EC3h. These registers are written to zero upon power up. From EC0h to EC3h, the name of these registers are **Port 0 data**, **Port 1 data**, **Multi-Function**, and **DIO-Control** register.



Note While the 16-pin aDIO connector is pin-for-pin compatible with previous generations of RTD cpuModules (such as the Montevina series), the hardware is different, and requires that writes and reads to/from the aDIO ports happen at a 16.66KHz rate (60usec delay between R/W operations).

Note RTD provides drivers that support the aDIO interface on popular operating systems. RTD recommends using these drivers instead of accessing the registers directly.

Digital I/O Register Set

Table 34 Port 0 Data I/O Address EC0h

D7	D6	D5	D4	D3	D2	D1	D0
P0.7	P0.6	P0.5	P0.4	P0.3	P0.2	P0.1	P0.0

Port 0 Data register is a read/write bit direction programmable register. A particular bit can be set to input or output. A read of an input bit returns the value of port 0. A read of an output bit returns the last value written to Port 0. A write to an output bit sends that value to port 0.

Table 35 Port 1 Data I/O Address EC1h

D7	D6	D5	D4	D3	D2	D1	D0
Reserved	Reserved	Reserved	Reserved	P1.3	P1.2	P1.1	P1.0

Port 1 Data register is a read/write byte direction programmable register. A read on this register when it is programmed to input will read the value at the aDIO connector. A write on this register when it is programmed as output will write the value to the aDIO connector. A read on this register when it is set to output will read the last value sent to the aDIO connector.

Table 36 Multi-Function I/O Address EC2h

D7	D6	D5	D4	D3	D2	D1	D0
----	----	----	----	----	----	----	----

The multi-function register is a read/write register whose contents are set by the DIO-Control register. See the DIO-Control register description for a description of this register.

Table 37 DIO-Control I/O Address EC3h—Read Access

D7	D6	D5	D4	D3	D2	D1	D0
Strobe 0 Status 0 = no strobe		Strobe 1 Status 0 = no strobe		Digital IRQ Mode 00 = Disabled 01 = strobe Mode 10 = event mode 11 = match mode		Port 1 Direction 0 = input	
Digital IRQ Status 0 = no digital interrupt 1 = digital interrupt						Multi-Function Register Select 00 = clear mode 01 = port 0 direction 10 = mask register 11 = compare register	

Table 38 DIO-Control I/O Address EC3h—Write Access

D7	D6	D5	D4	D3	D2	D1	D0
Reserved				Digital IRQ Mode 00 = Disabled 01 = strobe Mode 10 = event mode 11 = match mode		Port 1 Direction 0 = input	
						Multi-Function Register Select 00 = clear mode 01 = port 0 direction 10 = mask register 11 = compare register	

Table 39 Multi-Function at Address EC2h ¹									
read/write	00 clear	X	X	X	X	X	X	X	X
		I/O	I/O	I/O	I/O	I/O	I/O	I/O	I/O
0 in, 1 out	01 Port Direction	M7	M6	M5	M4	M3	M2	M1	M0
		C7	C6	C5	C4	C3	C2	C1	C0
0 no mask, 1 mask	10 DIO mask								
read/write	11 compare								

1. Contents based on bits D0 and D1 of DIO-Control.

Clear Register:

A read to this register Clears the IRQs and a write to this register sets the DIO-Compare, DIO- Mask, DIO-Control, Port 1, and Port 0 to zeros. A write to this register is used to clear the board.

Port 0 Direction Register:

Writing a zero to a bit in this register makes the corresponding pin of the aDIO connector an input. Writing a one to a bit in this register makes the corresponding pin of the aDIO connector an output.

Mask Register:

Writing a zero to a bit in this register will not mask off the corresponding bit in the DIO-Compare register. Writing a one to a bit in this register masks off the corresponding bit in the DIO-Compare register. When all bits are masked off the aDIOs comparator is disabled. This condition means Event and Match mode will not generate an interrupt. This register is used by Event and Match modes.

Compare Register:

A Read/Write register used for Match Mode. Bit values in this register that are not masked off are compared against the value on Port 0. A Match or Event causes bit 6 of DIO-Control to be set and if the aDIO is in Advanced interrupt mode, the Match or Event causes an interrupt. Port 1 Data register is a read/write byte direction

Interrupts

Interrupt servicing is code and OS dependent. To use an interrupt with aDIO, the interrupt must first be selected in the BIOS setup utility under **Advanced, RTD Advanced Options, RTD aDIO, aDIO Interrupt**. The Digital I/O can use interrupts 3, 5, 6, 10, and 11. To configure the aDIO interrupt, navigate to the BIOS Setup option, the “Advanced”, “RTD aDIO”, and “aDIO Interrupt” menu options. The BIOS will automatically reserve the selected interrupt so that it is not assigned to PCI devices. Then, select the appropriate interrupt mode in the DIO Control register.

Advanced Digital Interrupts

There are three Advanced Digital Interrupt modes available. These three modes are Event, Match, and Strobe. The use of these three modes is to monitor state changes at the aDIO connector. Interrupts are enabled by writing to the **Digital IRQ Mode** field in the **DIO-Control** register.

Event Mode

When this mode is enabled, Port 0 is latched into the DIO-Compare register. The aDIO circuitry includes deglitching logic. To enter Event mode, set bits [4:3] of the DIO-Control register to “10”.

Match Mode

When this mode is enabled, Port 0 is latched into the DIO-Compare register. The aDIO circuitry includes deglitching logic. To enter Match mode, set bits [4:3] of the DIO-Control register to “11”.



Note Make sure bits [4:3] are set BEFORE writing the DIO-Compare register. If you do not set them first, the contents of the DIO-Compare register could be lost because the Event mode latches in Port 0 into the DIO-Compare register.

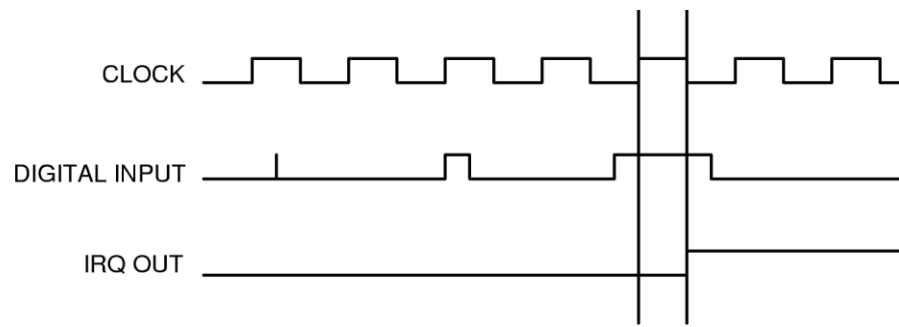


Figure 7 aDIO Match Mode

Strobe Mode

Another interrupt mode supported by aDIO is Strobe mode. This allows the strobe pin of the DIO connector to trigger an interrupt. A low to high transition on the strobe pin will cause an interrupt request. The request will remain high until the Clear Register is read from. Additionally, the Compare Register latched in the value at Port 0 when the Strobe pin made a low to high transition. No further strobes will be available until a read of the Compare Register is made. You must read the Compare Register, and then clear interrupts so that the latched value in the compare register is not lost. To enter Strobe mode, set bits [4:3] of the DIO-Control register to “01”.

SATA Controller Configuration

The CMX34KB/CMA34KB has six SATA 3.0 links with transfer speeds up to 6 Gbps. Two SATA links reside on the top PCIe/104 Type 2 connector (**CN1**), and two SATA links reside on the bottom PCIe/104 Type 2 connector (**CN2**). The remaining SATA links connect to the onboard SATA storage drive and to the SATA connector, **CN15**. Both links may be enabled or disabled through the BIOS setup.

SATA Port Mappings, by Connector

Table 38 shows the SATA port configuration of the CMX34KB/CMA34KB cpuModule:

Table 40 SATA Port Layout

Designator	Physical Position, Index	Chipset Port
CN1	SATA PCIe link 2	2
CN1	SATA PCIe link 3	3
CN2	SATA PCIe link 4	4
CN2	SATA PCIe link 5	5
CN15	SATA Data Connector	1
n/a	Onboard SATA storage	0

Onboard SATA Storage

The cpuModule was designed to be used in embedded computing applications. In such environments, rotating media like hard disks and floppy disks are not very desirable. It is possible to eliminate rotating storage devices by placing your operating system and application software into the cpuModule's onboard SATA storage.

The onboard SATA storage supports write protection. Write protection is disabled by default and can be enabled with a menu option in the BIOS setup. If the onboard SATA storage contains an OS that requires writing to function normally, the user may see their system perform abnormally.

SATA Links on the PCIe/104 Type Connectors

Four of the six SATA links on the CMX34KB/CMA34KB are connected to the CPU's SATA controller which is configured for ACHI through the BIOS. Two links go to the top-side PCIe/104 connector, and two links go to the bottom-side PCIe/104 connector.

Watchdog Timer Control

The cpuModule includes a Watchdog Timer, which provides protection against programs “hanging”, or getting stuck in an execution loop where they cannot respond correctly. The watchdog timer consists of a counter, a reset generator, and an interrupt generator. When the counter reaches the interrupt time-out, it can generate an interrupt. When the counter reaches the reset time-out, the system is reset. The counter is “refreshed,” or set back to zero by reading from a specific register. The watchdog can also be put into an “inactive” state, in which no resets or interrupts are generated.

The ability to generate an interrupt allows the application to gracefully recover from a bad state. For example, consider a system that has a reset time-out of 2 seconds, interrupt time-out of 1 second, and the watchdog timer is refreshed every 0.5 seconds. If something goes wrong, an interrupt is generated. The Interrupt service routine then attempts to restart the application software. If it is successful, the application is restarted in much less time than a full reboot would require. If it is not successful, the system is rebooted.

Due to system latency, it is recommended that the Watchdog be refreshed at about half of the reset time-out period, or half of the interrupt time-out period, whichever is applicable.

Register Description

The Advanced Watchdog Timer has a Setup Register and a Runtime Register. The Setup Register is set by the BIOS and can be adjusted by entering the BIOS Setup Utility, and going to “Advanced”, “RTD Advanced Options”, “RTD Miscellaneous Feature Control”. The Setup Register may also be read by the driver to determine if the Watchdog is enabled, and the interrupt and base address that it is using.



Note Enabling the watchdog timer in the BIOS does not actually arm it. The watchdog timer can be armed by accessing I/O address E85h, as explained below.

Table 41 Advanced Watchdog Setup Register E8Bh

D7	D6	D5	D4	D3	D2	D1	D0
Reserved				WDT_IRQ		Reg_Enable	
				Select Interrupt for WDT		0 = Watchdog timer is disabled and Runtime Register will not appear in I/O map	
				000 = Disabled		1 = Watchdog Timer is enabled.	
				001 = IRQ5		Runtime Register will appear in I/O map	
				010 = Reserved			
				011 = IRQ10			
				100 = IRQ11			
				101 = Reserved			
				110 = IRQ3			
				111 = IRQ6			

Table 42 Advanced Watchdog Runtime Register E85h

D7	D6	D5	D4	D3	D2	D1	D0
WDT_Active	WDT_IRQ_En	Reserved		WDT_IRQ_Ti		WDT_RST_Ti	
0 = Watchdog timer is disabled.	0 = Watchdog interrupt is disabled.			Select Interrupt time for WDT		Select Reset time for WDT	
1 = Watchdog is armed and can generate resets and interrupts.	1 = Watchdog interrupt is enabled.			00 = 0.25s		00 = 2.00s	
				01 = 0.50s		01 = 0.50s	
				10 = 0.75s		10 = 0.75s	
				11 = 1.00s		11 = 1.00s	

Reading the Runtime Register also refreshes the watchdog timer.

Thermal Management

The cpuModule has several thermal features which can be used to monitor and control the board's temperature when extreme operating conditions are prevalent.

Thermal Warning

The cpuModule includes a Thermal Warning mechanism which will assert a signal when the processor's temperature is at or above 100 C. At this time, a bit will be set in the reset and event status register to indicate that this temperature has been reached. Although the LED has turned white and a thermal warning bit has been set, the CPU will maintain the current power and operating state.

For more information on the reset status register, refer to *Reset and Event Status Register* – Page 74.

ACPI-Enforced Thermal Protection

When the maximum thermal junction temperature of the processor is reached, the system is at risk of malfunctioning, and the cpuModule may become permanently damaged.

For operating system with ACPI support, the system offers two mechanisms to help prevent the processor from reaching its maximum thermal junction temperature -- the **Critical Trip Point** and the **Passive Trip Point**. Both trip point temperatures may be configured in the BIOS Setup.

Critical Trip Point

The Critical Trip Point controls the temperature at which an ACPI operating system will shut down the system. Upon entering the Critical Trip state, the operating system will issue a command to the processor, and the CMX34KB/CMA34KB will immediately transition to the S5 (Soft-Off) ACPI power state.

To exit the Critical Trip state (and S5), the CPU temperature must be reduced to a point that the internal thermal sensors are below the Critical Trip Point. Next, a power cycle or RSM reset (deepest reset) is required. The reset button may be configured as an RSM reset (default BIOS setting) in the BIOS setup utility.

For more information on the S5 power state, refer to *Advanced Configuration and Power Interface (ACPI)* – Page 71.

Passive Trip Point

The Passive Trip Point controls the temperature at which an ACPI operating system will begin to throttle the processor.

Further Temperature Reduction

The cpuModule's temperature is directly related to power consumption. Reducing the power consumption of the cpuModule will affect its temperature. Suggested methods for reducing the cpuModule's power consumption can be found in the *Power Management* section on page 71.

Power Management

The CMX34KB/CMA34KB cpuModule supports various powering mechanisms which allow the cpuModule to monitor power consumption and temperature and achieve minimal power consumption states. These unique features include thermal monitoring and thermal throttling, as well as ACPI low power modes. Various wake options are also available to resume normal system power.

Advanced Configuration and Power Interface (ACPI)

The cpuModule supports several different ACPI low power modes, including the S3, S4, and S5 sleeping states. These suspend modes are described below:

- **S3 (Suspend to RAM):** When the system wakes from this mode, operating systems allow applications to resume where they left off, as the state of the application is preserved in memory.
- **S4 (Hibernate):** When the system enters this state, the operating system will save the current state of applications and relevant data to disk, thus allowing the system RAM to be powered down.
- **S5 (Soft-Off):** The system is in a soft off state and must be rebooted when it wakes.

Power Button Modes

The soft power button input of the Utility Port 2.0 connector (CN5) can be configured by the operating system as a suspend button (transition to S3) or as soft power button (transition to S5). Consult your operating system documentation for information on how to configure it. The power button will always cause a transition to S5 if pressed for 4 seconds or longer, without interaction from the operating system.

The soft power button of the Utility Port 2.0 connector (CN5) is the only mechanism to wake the system from S4 and S5.



Note When the reset button of the Utility Port 2.0 connector is configured as a power button (via the BIOS setup utility), it inherits the ACPI suspend and resume features of the power button.

Low-Power Wake Options

The cpuModule supports several methods of waking from a low power state.

- **Resume from USB:** Operating systems that support S3 permit waking the system from the USB interfaces. Common USB wake methods include insertion of a USB device, a USB keyboard stroke, or movement from another USB device, such as mouse.
- **Resume on RTC Alarm / Timeout:** The RTC Alarm allows the system to turn on at a certain time every day.

Table 43 Supported ACPI Resume/Wake Mechanisms

Resume/Wake Mechanism	S3	S4	S5
Wake from USB	Y	—	—
Wake from RTC Alarm / Timeout	Y	—	—
Power Button Input (CN5)	Y	Y	Y

AT vs. ATX Power Supplies

Both AT and ATX power supplies may be used with the CMX34KB/CMA34KB cpuModule, however AT power supplies do not provide any standby power to the cpuModule. When an AT power supply is used to power the system, low power modes that require a standby power to wake the system will not be fully supported.

ATX power supplies do provide a standby power, thus allowing the system to utilize all low power modes supported by the hardware. When an ATX supply is used to power the cpuModule, lower power modes can be achieved. During these low power modes, the standby power from the ATX power supply provides power to a small circuit on the CPU, which is used to watch for a system wake event.

ATX Power Supply Signals

The PCIe/104 Type 2 Bus connectors (**CN1** & **CN2**) provide two ATX style signals., +5V Standby and PSON#. The +5V Standby rail is used to power certain parts of the cpuModule when the main power supply is turned off, i.e., during Suspend-to-RAM (S3), Hibernate (S4), or Soft-Off (S5) power modes. The PSON# signal is an active low open-drain output that signals the power supply to turn on. Use of these signals allows the power consumption to drop to below 1W during standby modes, and still enable any of the wake events.

Reducing Power Consumption

In addition to the CPU's low power modes, power consumption can further be reduced by making some modifications to the BIOS setup. When the following features are disabled in the BIOS, the CPU's power consumption will decrease:

- **Ethernet**
- **USB Ports**
- **SATA Controller**
- **Serial Ports**
- **Multi-Color LED**

Multi-Color LED

The CMX34KB/CMA34KB has a Multi-Color LED which can be enabled or disabled in the BIOS setup screen. The color of the LED indicates the status of the board, as shown in Table 42.

Table 44 LED Colors

Color	Description
Green	Normal Operation
Blue	SATA Activity
Red ¹	cpuModule is in reset
Yellow (Red + Green)	cpuModule is in Standby
White (R+G+B)	cpuModule is approaching thermal limit ²
Cyan (Blue + Green)	Ethernet Link at 100 Mbps or Bus Stacking Error
Magenta (Blue + Red)	Ethernet Link at 1000 Mbps
Blink	Ethernet Activity

1. If power is applied to the cpuModule while jumper **JP5** is installed, the LED will be red. This does not indicate that the board is in reset
2. The LED will remain White until the system is shut down.

The LED can also be controlled manually by writing to I/O Port EA7h, as shown in Table 43 and Table 44.

Table 45 Multi-Color LED I/O Address EA5h

D7	D6	D5	D4	D3	D2	D1	D0
Reserved	Reserved	Reserved	Reserved	Reserved		Multi-Color LED	

The following table lists the color displayed and the value written.

Table 46 Manual LED Colors

I/O Port EA7h Value	Color
0x00	Automatic (see Table 42)
0x08	Off (will reduce system power consumption)
0x09	Blue
0x0A	Green
0x0B	Cyan (Green + Blue)
0x0C	Red
0x0D	Magenta (Red + Blue)
0x0E	Yellow (Red + Green)
0x0F	White (Red + Green + Blue)

Reset and Event Status Register

The cpuModule has several different signals on board which can cause a system reset. If a reset occurs, the reset status register can be used to see which reset (or resets) have been asserted on the cpuModule. These resets may be cleared by the user.

Like resets, there are several signals on the cpuModule which indicate that a special event has occurred. These may also be monitored and cleared by the user.

- **Examine Resets and Events:** Reading from I/O ports 0xED0h through 0xED3 will indicate if a reset has been asserted or if an event has occurred. If a 1 is read, the corresponding reset has been asserted. If a 0 is read from the bit, the reset has not been asserted.
- **Clear Reset and Event:** Each reset and event can be cleared by writing a 1 to the selected bit of I/O port 0xED0h.

Table 47 Reset and Event Status I/O Address ED0h - Read Access

D7	D6	D5	D4	D3	D2	D1	D0
Sleep S4 1 = reset asserted 0 = no reset	PwrGood 1.8V Always 1 = reset asserted	PwrGood 1.0V Always 1 = reset asserted 0 = no reset	PwrGood 5V Always 1 = reset asserted	RSM Reset 1 = reset asserted 0 = no reset	PwrGood 1.5V Always 1 = reset asserted	PwrGood 3.3V Always 1 = reset asserted 0 = no reset	Main 5V Power 1 = reset asserted 0 = no reset

Table 48 Reset and Event Status I/O Address ED1h - Read Access

D7	D6	D5	D4	D3	D2	D1	D0
PwrGood 1.05V 1 = reset asserted 0 = no reset	Pwr Good Vcore 1 = reset asserted 0 = no reset	Sleep S3 1 = reset asserted 0 = no reset	PwrGood 1.2V Always 1 = reset asserted	PwrGood 1.0V 1 = reset asserted 0 = no reset	PwrGood Vgfx 1 = reset asserted 0 = no reset	PwrGood 1.9V Always 1 = reset asserted 0 = no reset	Memory Pwr Good 1 = reset asserted

Table 49 Reset and Event Status I/O Address ED2h - Read Access

D7	D6	D5	D4	D3	D2	D1	D0
WDT Trip 1 = reset asserted 0 = no reset	Thermal Warning 1 = reset asserted	Platform Reset 1 = reset asserted 0 = no reset	Reserved	Mem. Term. Voltage 1 = reset asserted 0 = no reset	PwrGood 3.3V 1 = reset asserted 0 = no reset	PwrGood 1.8V 1 = reset asserted 0 = no reset	PwrGood 1.35V 1 = reset asserted 0 = no reset

Table 50 Reset and Event Status I/O Address ED3h - Read Access

D7	D6	D5	D4	D3	D2	D1	D0
Reserved	Reserved	ATX Pwr Good 1 = reset asserted 0 = no reset	Reserved	Reserved	Soft Power Off 1 = reset asserted 0 = no reset	Utility Reset 1 = reset asserted 0 = no reset	Reserved
			Bus Stack Error 1 = reset asserted 0 = no reset				

Trusted Platform Module

The CMX34KB/CMA34KB cpuModule includes a Trusted Platform Module (TPM) that is fully compliant to the Trusted Computing Group (TCG) TPM version 2.0 specification.

TPM stands for “Trusted Platform Module”. The TPM is a controller on your cpuModule that provides tamper resistant storage using hardware-based cryptography. The TPM can be used to store passwords, certificates, and keys to encrypted file systems. The TPM protects the data on your drives from being read should your system or drive be stolen. An un-encrypted drive can be removed from your system and connected to another, bypassing the need for a user account and password. With a TPM encrypted drive the only way to access the data is by logging into the system as a user with access to the files containing your sensitive information. TPM 2.0 controllers can generate and store [AES-256-GCM-Sha256](#) encryption keys to ensure your data can not be decrypted. As such, it is highly recommended to save a copy of any file system encryption keys either on paper or a USB drive in the event your device experiences a hardware failure. Without this key your data will be lost with NO chance of recovery. For best results, this key should never be saved locally, even temporarily, as the key may leave a lasting trace on flash or magnetic drives.

TPM chips are still vulnerable to software attacks, malware, and viruses. Once the system has booted and a user is logged in, the encrypted drives are unlocked with the keys from the TPM, allowing any software attacks full access to any important data and even the keys within the TPM. As such, it is still highly recommended to maintain firewalls, anti-virus, and any other internet security methods.

Before using the TPM chip it must be enabled and cleared in the Security Tab of the BIOS. Windows must also be installed in UEFI mode to allow TPM access. UEFI boot mode can be set in the Boot tab of the bios. For Debian-based Linux distributions refer to “TPM Implementation White Paper”. For Red Hat Linux refer to guides found on the Red Hat Customer Portal for your version.

For more information, refer to the Trusted Platform Module version 2.0 specification.

Table 51 Additional References:

Title	Description	Source
TPM Specifications	TPM specs, versions, updates	https://trustedcomputinggroup.org/resource/tpm-library-specification/
TPM Summary and Operation	An in-depth white paper on how the TPM works.	https://trustedcomputinggroup.org/resource/trusted-platform-module-tpm-summary/
TPM Implementation White Paper	An in-depth white paper on how to set up TPM drivers in Windows and Linux	Contact RTD Techsupport
Red Hat Customer Portal	Customer portal for Red Hat Subscribers	https://access.redhat.com/

Features and Settings That Can Affect Boot Time

The boot time of a system is dependent upon numerous system settings as well as devices attached to a system. This section addresses some devices and settings that can increase or decrease a system's boot time.

Boot Device Order

The BIOS contains a list of devices to try booting from. If you wish to boot to a particular device (for example, a hard drive), make sure that it is first in the boot order. This will speed up boot times.

Add-On Cards with BIOS Extensions

Some add-on cards have an integrated BIOS extension. The most common examples are SCSI controllers and network cards with boot ROMs. During POST, the BIOS executes the card's extension code. This extension code is third-party code, which is beyond RTD's control. The BIOS extension will most likely increase the boot time. Exactly how much it increases boot time will depend on the particular card and firmware version.

Hard Drive Type

During Hard Drive initialization, each device must be probed. Some devices take longer to probe. 2.5-inch hard drives tend to take longer than 3.5-inch ones, because they spin at a lower RPM.

Monitor Type

Some monitors take a while to power on. Desktop flat panels are especially slow. This does not affect the actual boot time of the CPU. However, the CPU may boot before the monitor powers on.

System Recovery

Reset Button Recovery

The CMX34KB/CMA34KB provides several methods for recovering from an incorrectly configured system. To enter the recovery mode, follow the steps below:

1. Remove power from the system, including standby power.
2. Press and hold the reset button attached to the Utility Port 2.0 connector.
3. Apply power to the system while continuing to hold the reset button.
4. Wait the amount of time shown in Table 50 for the desired recovery mode.
5. Release the reset button, allowing the system to boot.

Table 52 Reset Button Recovery Modes

Hold Time ¹	Mode
0-4 seconds	No recovery mode. System stays in reset while button is pressed.
4-8 seconds	Load Default BIOS Settings
8-12 seconds	Serial Post Code Output

1. To assist with timing while the reset button is pushed, the LED will blink OFF at 5 second intervals.

Load Default BIOS Settings

Loading BIOS defaults allows recovery from an incorrectly configured display device, incorrect boot options, and many other incorrect settings. It is also a good starting point when making BIOS changes. After restoring defaults, the BIOS settings should be reviewed and modified as needed.

The default BIOS can be restored either by using Reset Button Recovery, or the “Load RTD Defaults” option in the BIOS.

Serial Power-On-Self-Test (POST) Code Output

The POST Codes represent a series of events that take place in a system during the Power-On Self-Test. If the POST fails, the system will not boot as expected. The POST code immediately before the failure occurs may help with system debug.

This recovery mode configures serial port connector **CN7** as single RS-232 and sends the POST codes on the transmit pin. The port settings are 115kbps, 8 bits, no parity, one stop bit. When using this recovery mode, the POST codes can be logged on another computer running terminal software. Contact RTD technical support for more details.



Note POST Codes over the serial port (**CN7**) may also be enabled by installing **JP5** and **JP6** prior to power cycling the cpuModule. For more information, see Jumper Settings and Locations on page 80.

Appendix A Hardware Reference

This appendix provides information on CMX34KB/CMA34KB cpuModule hardware, including:

Jumper Settings and Locations – Page 80

Onboard PCI/PCIe Devices – Page 81

Physical Dimensions – Page 82

Heatsink Dimensions – Page 83

Flat-Heatspreader Dimensions – Page 84

Jumper Settings and Locations

Many cpuModule options are configured by positioning jumpers. Jumpers are labeled on the board as **JP** followed by a number.

Figure 8 shows the jumper locations that are used to configure the cpuModule. Table 51 lists the jumpers and their settings. Both the CMX34KB and the CMA34KB have the same jumpers and jumper settings.

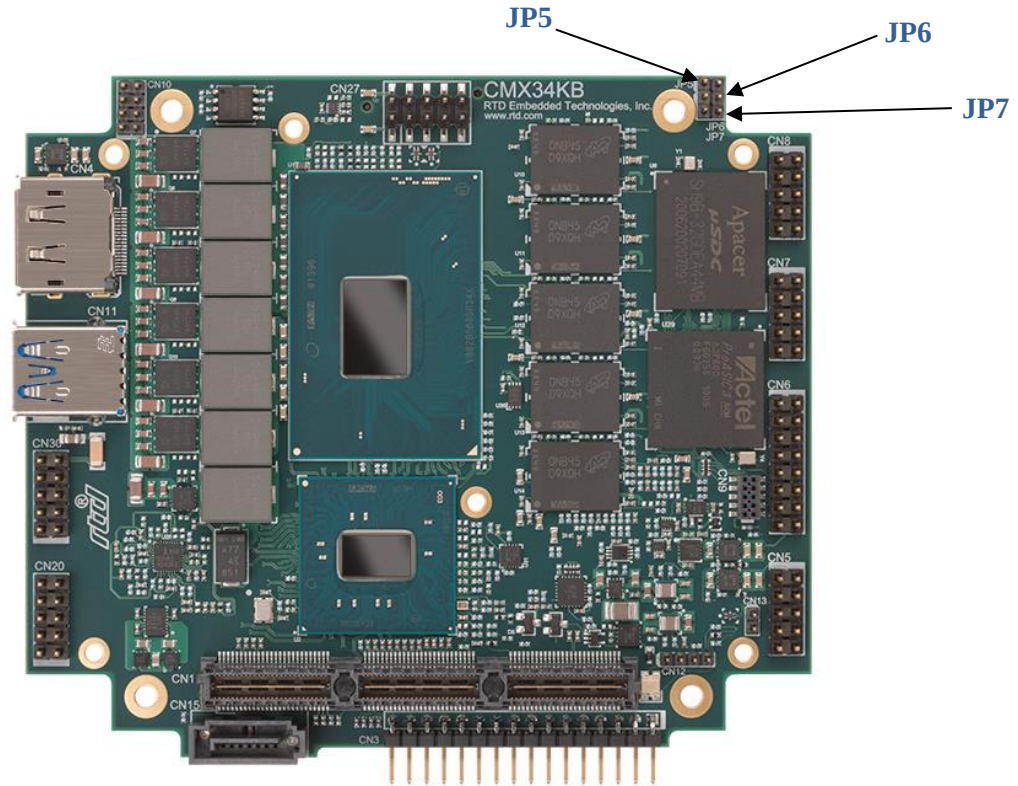


Figure 8 CMX34KB/CMA34KB Jumper Locations (top side)

Table 53 CMX34KB/CMA34KB Jumpers

Jumper	Pins	Function	Default
JP5	2	Used to load the default BIOS settings. Install this jumper and boot the stack to load the RTD default BIOS settings, remove the jumper once the system has booted.	open
JP6	2	Used to disable the Bus Stacking Error detection. See PCIe/104 Type 2 Compatibility on page 54. pins 1–2: Disable Bus Stacking Error detection open: Enable Bus Stacking Error detection (normal operation)	open
JP5 + JP6	2	Used to configure serial port connector CN7 as single RS-232 and sends the POST codes on the transmit pin. See Serial Power-On-Self-Test (POST) Code Output on page 78. Pins 1–2 (JP5 and JP6): Enable POST Codes over Serial Port CN7 open: Normal operation.	open
JP7	2	Reserved1	open

1. If JP7 is shorted the cpuModule will not boot.

Onboard PCI/PCIe Devices

The CMX34KB/CMA34KB cpuModule has several onboard PCI/PCIe devices, all of which are listed in the table below. This table shows a typical configuration, and the actual devices may change based on BIOS settings.

Table 54 Onboard PCI/PCIe Devices

Device ID	Vendor ID	Description
5918	8086	Host bridge
591D	8086	VGA-Compatible Controller
1911	8086	Other system Peripheral (Xeon E3-1200)
A12F	8086	Universal Serial Bus (USB) xHCI
A131	8086	100 Series/C230 Series Chipset Family Thermal Subsystem
A160	8086	100 Series/C230 Series Chipset Family Serial IO I2C Controller #0
A161	8086	100 Series/C230 Series Chipset Family Serial IO I2C Controller #1
A13A	8086	100 Series/C230 Series Chipset Family MEI Controller #1
A102	8086	SATA Controller
A169	8086	PCI-to-PCI bridge
A16A	8086	PCI-to-PCI bridge
A154	8086	ISA bridge
A121	8086	Other memory controller
A171	8086	HDA Audio Controller
A123	8086	SMBus (System Management Bus)
1533	8086	Ethernet Controller - Intel WGI210IT Series
1533	8086	Ethernet Controller - Intel WGI210IT Series

Physical Dimensions

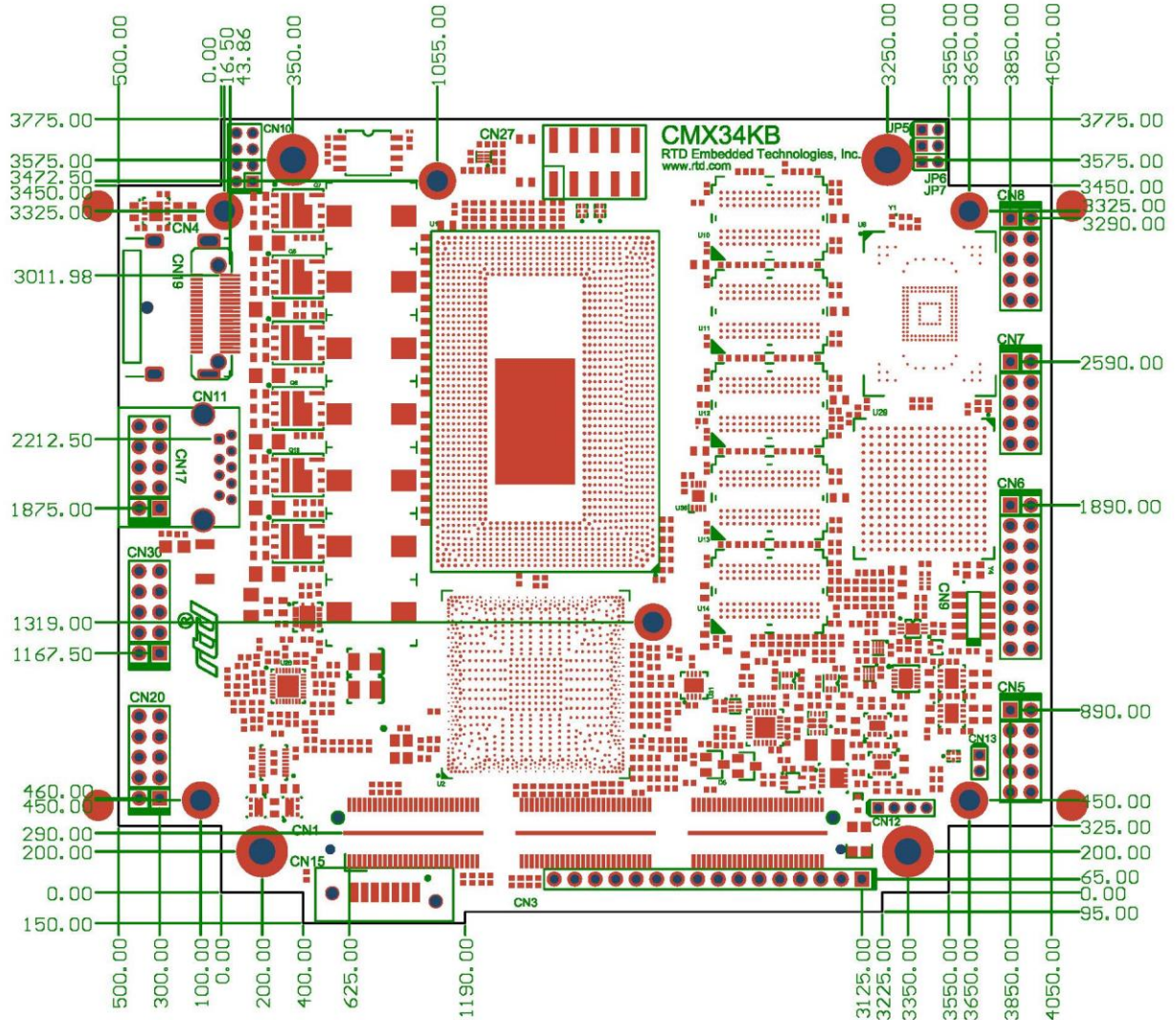


Figure 9¹ CMX34KB/CMA34KB Physical Dimensions (± 0.005 inches)

1. Both CMX34KB and CMA34KB have the same dimensions

Heatsink Dimensions

To maximize the thermal performance of the cpuModule while keeping the total board weight to a minimum, various heatsinks are offered with each ordering part number. Use the table below to correlate the heatsink figure to each ordering part number.

Table 55 CMX34KB/CMA34KB cpuModule Model Options

Part Number	Core Frequency	Stack Height	Heatsink
CMX34KBD2100HR-16384/S60GX	2.1 GHz	0.866" (22mm)	Figure 10
CMX34KBQ2200HR-16384/S60GX	2.2 GHz	0.866" (22mm)	Figure 10
CMX34KBQ3000HR-16384/S60GX	3.0 GHz	0.866" (22mm)	Figure 10
CMA34KBD2100HR-16384/S60GX	2.1 GHz	0.866" (22mm)	Figure 11
CMA34KBQ2200HR-16384/S60GX	2.2 GHz	0.866" (22mm)	Figure 11
CMA34KBQ3000HR-16384/S60GX	3.0 GHz	0.866" (22mm)	Figure 11

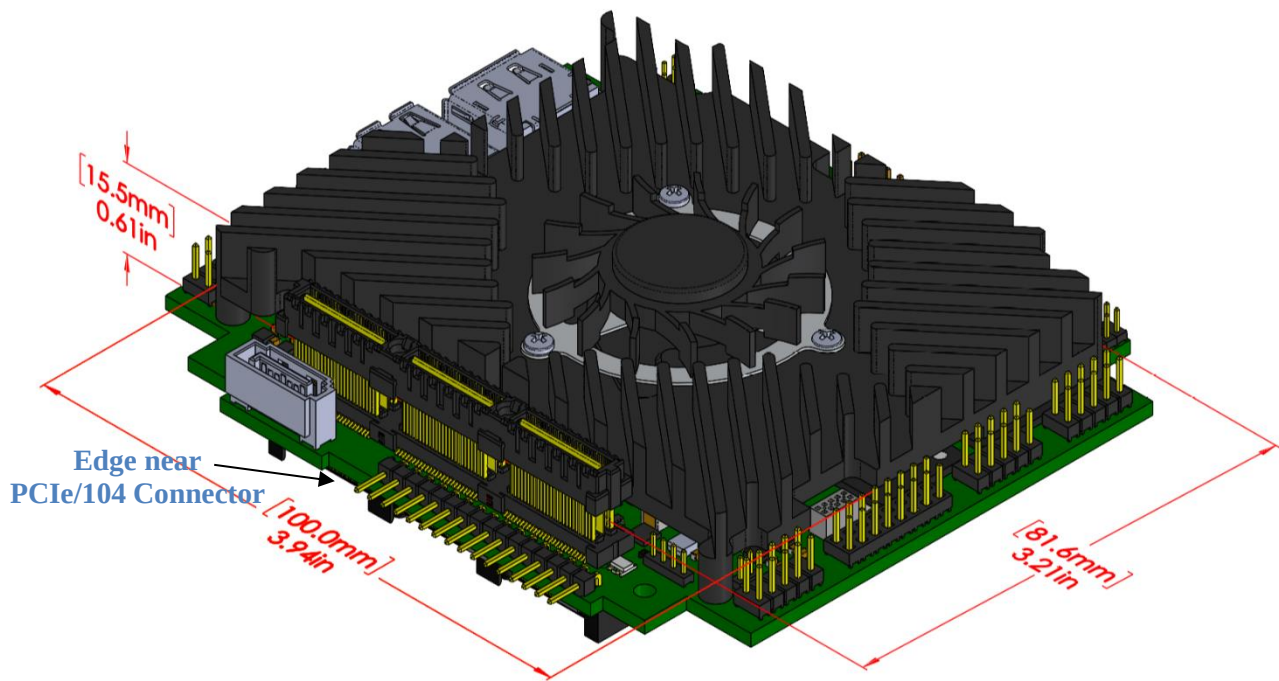


Figure 10 CMX34KB Heatsink

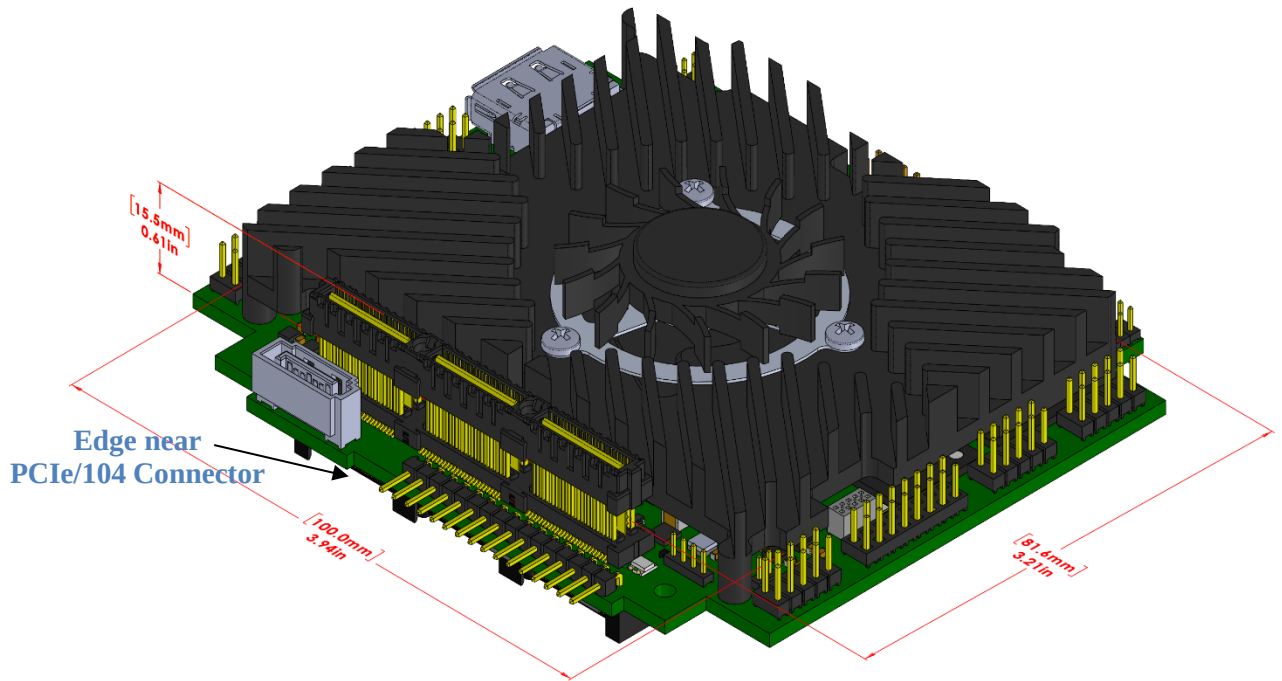


Figure 10 CMA34KB Heatsink

Flat-Heatspreader Dimensions

For cooling solutions where it is desirable to mount the cpuModule to a flat surface, the CMX34KB/CMA34KB may be ordered with a flat heatspreader instead of the standard passive heatsink. Each flat-heatspreader has five mounting locations which use 6-32 UNC x 0.207" long threaded insert.

Use the table below to correlate the heatsink figure to each ordering part number.

Table 56 CMX34KB/CMA34KB cpuModule Model Options

Part Number	Core Frequency	Stack Height	Heatspreader
CMX34KBD2100HR-16384/S60GXF	2.1 GHz	0.866" (22mm)	Figure 13
CMX34KBQ2200HR-16384/S60GXF	2.2 GHz	0.866" (22mm)	Figure 13
CMX34KBQ3000HR-16384/S60GXF	3.0 GHz	0.866" (22mm)	Figure 13
CMA34KBD2100HR-16384/S60GX	2.1 GHz	0.866" (22mm)	Figure 14
CMA34KBQ2200HR-16384/S60GXF	2.2 GHz	0.866" (22mm)	Figure 14
CMA34KBQ3000HR-16384/S60GXF	3.0 GHz	0.866" (22mm)	Figure 14

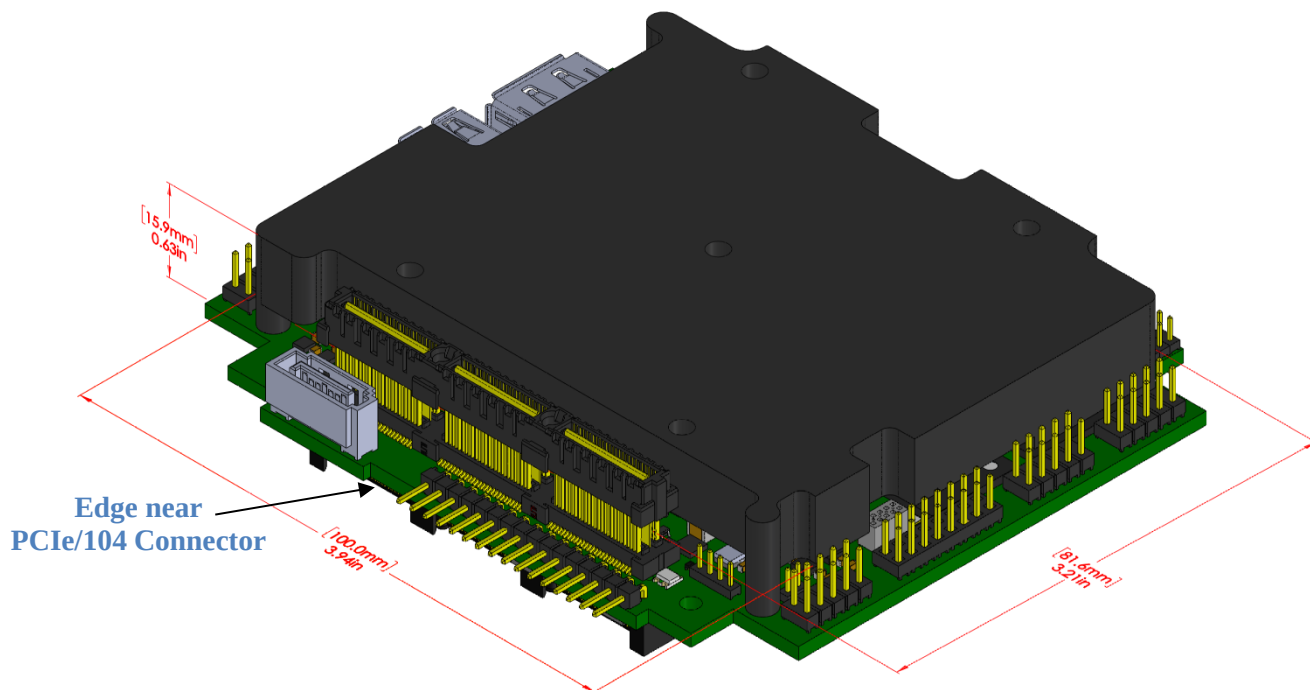


Figure 13 CMX34KB 0.866" (22mm) Flat-Heatspreader

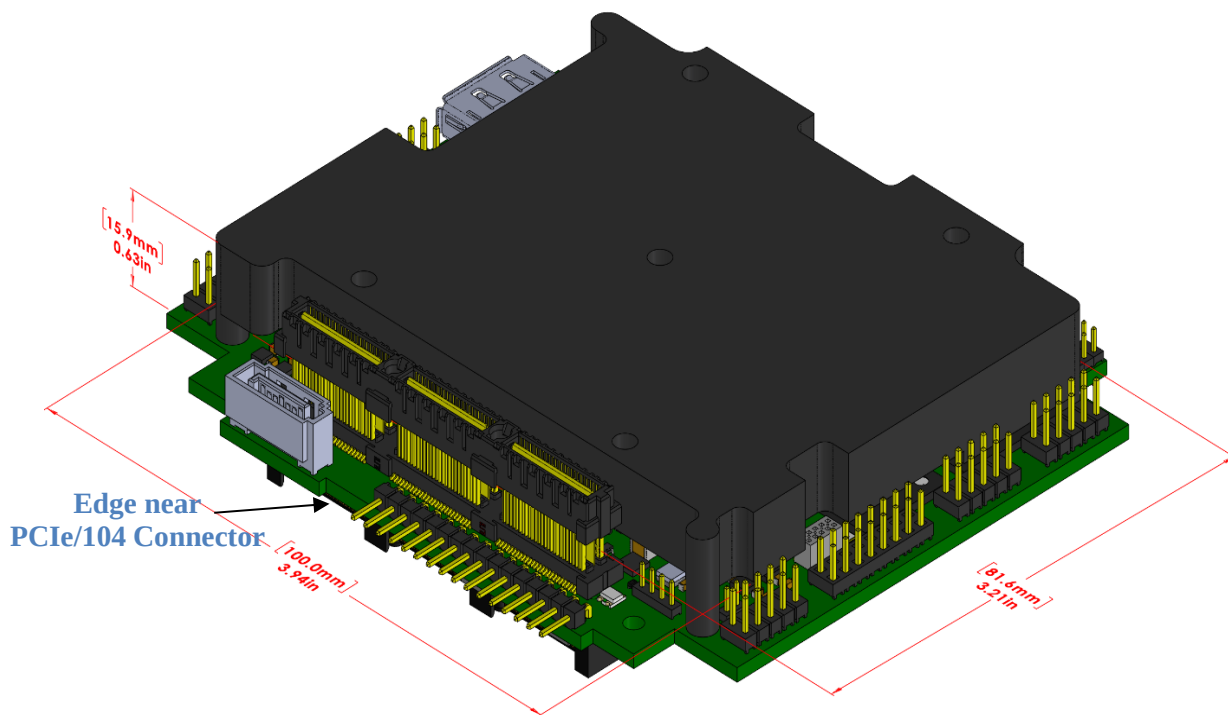


Figure 14 CMA34KB 0.866" (22mm) Flat-Heatspreader

Appendix B Troubleshooting

Many problems you may encounter with operation of your CMX34KB/CMA34KB cpuModule are due to common errors. This appendix includes the following sections to help you get your system operating properly.

Common Problems and Solutions – Page 87

Troubleshooting a PC/104 System – Page 88

How to Obtain Technical Support – Page 89

Common Problems and Solutions

Table 55 lists some of the common problems you may encounter while using your CMX34KB/CMA34KB cpuModule and suggests possible solutions.

If you are having problems with your cpuModule, review this table before contacting RTD Technical Support.

Table 57 Troubleshooting

Problem	Additional Symptoms	Possible Cause(s)
cpuModule does not boot (no video output)	LED is red (in reset)	Input power is inadequate. Refer to Chapter 1, <i>Electrical Characteristics</i> . Verify input voltages are correct. If power is supplied via CN3 , check for cabling issues. Malfunctioning peripheral card. Remove all peripheral cards from the system.
	LED is yellow (in standby)	cpuModule is in “Soft Off” due to operating system shutdown. Press power button to wake. ATX power supply is not turning on. Check jumpers on power supply. Verify PSON# signal.
	LED is cyan (bus stacking error)	See Chapter 3
	LED is green (normal operation)	cpuModule may be booted, but video is not connected properly. Attach a keyboard and verify the operation of the Num Lock and Caps Lock LEDs. If they respond as expected, the cpuModule is booted. Check the video cable connections. Video output may have been disabled in the BIOS. Try loading BIOS defaults per Chapter 4, <i>System Recovery</i> . If a PCI device is installed, the slot selection switch/jumpers may not be set correctly. Peripheral card may be interfering with boot sequence. Remove all peripheral cards from the system.
cpuModule reboots unexpectedly	cpuModule is running Windows	By default, Windows will automatically reboot when a Bug check (Blue Screen) is triggered. Disable this feature in the Control Panel. This will allow you to see the error and identify the root cause.
	cpuModule is powered via Auxiliary Power Connector (CN3)	Power cabling issues. Verify all pins on CN3 are reliably connected with proper gauge wires. Power supply noise or current limiting. Check power pins with an oscilloscope and verify they remain in range. Power supply may not respond quickly enough to changes in power consumption. Switch to a different power supply (a PCI Express power supply is recommended)
	Peripheral card(s) installed	Peripheral card(s) may be overloading the power supply. Reduce the system to just the cpuModule and power supply and see if the problem re-appears. (Note that CN3 is intended for powering the cpuModule only.)
cpuModule does not boot to USB device	USB device has a high power draw (e.g., DVD or hard drive)	Some USB devices are not fully compliant and draw too much current. Consider switching to a self-powered USB device with its own AC/DC power supply.
Date/time not saved when system is off	No RTC battery	A battery must be attached to the Utility Port 2.0 connector (CN5) to maintain date/time when main power is removed. For a list of alternate battery inputs, refer to <i>Watchdog Timer Control</i> —page 69.

Table 58 Troubleshooting (cont'd)

Problem	Additional Symptoms	Possible Cause(s)
System performance lower than expected	LED is white	cpuModule is overheating and the processor is throttling. Increase the cooling (more airflow, larger heatsink, etc.).

Troubleshooting a PC/104 System

If you have reviewed the preceding table and still cannot isolate the problem with your CMX34KB/CMA34KB cpuModule, please try the following troubleshooting steps. Even if the resulting information does not help you find the problem, it will be very helpful if you need to contact technical support.

1. **Simplify the system.** Remove items one at a time and see if one item seems to cause the problem.
2. **Swap components.** Try replacing items in the system one-at-a-time with similar items.

How to Obtain Technical Support

If after following the above steps, you still cannot resolve a problem with your CMX34KB cpuModule, please gather the following information:

- cpuModule model, BIOS version, and serial number
- List of all boards in system
- List of settings from cpuModule Setup program
- Printout of autoexec.bat and config.sys files (if applicable)
- Description of problem
- Circumstances under which problem occurs

Then contact RTD Technical Support:

Phone: 814-234-8087

Fax: 814-234-5218

E-mail: techsupport@rtd.com

Appendix C IDAN[®] Dimensions and Pinout

Like all other RTD PC/PCI-104 and PCIe/104 modules, cpuModules can be packaged in Intelligent Data Acquisition Node (IDAN) frames, which are milled aluminum frames with integrated heat sinks and heat pipes for fan-less operation. RTD modules installed in IDAN frames are called building blocks. IDAN building blocks maintain the simple but rugged stacking concept of PC/PCI-104 and PCIe/104. Each RTD module is mounted in its own IDAN frame and all I/O connections are brought to the walls of each frame using standard PC connectors.

On the CMA34KB, no connections are made from module to module internal to the system other than through the PCIe/104 Type 2 bus, enabling quick interchangeability and system expansion without hours of rewiring and board redesign.

The CMA34KB cpuModule can also be purchased as part of a custom-built RTD HiDAN[®] or HiDANplus[®] High Reliability Intelligent Data Acquisition Node. This appendix provides the dimensions and pinouts of the CMX34KB/CMA34KB installed in an IDAN frame. Contact RTD for more information on high reliability IDAN, HiDAN, and HiDANplus PC/PCI-104 systems.

Note: The CMX34KB is not offered in an IDAN. We look to have an IDAN-CMX34KB in the future.





IDAN—Adhering to the PC/PCI-104 and PCIe/104 stacking concept, IDAN allows you to build a customized system with any combination of RTD modules.

IDAN Heat Pipes—Advanced heat pipe technology maximizes heat transfer to heat fins.



HiDANplus—Integrating the modularity of IDAN with the ruggedization of HiDAN, HiDANplus enables connectors on all system frames, with signals running between frames

IDAN Contents

The IDAN-CMA34KB contains both a CMA34KB cpuModule and a **SATA34106** storage module which provides a 2.5" SATA interface to the first SATA link on the CPU's top PCIe/104 Type 2 connector (**CN1**).

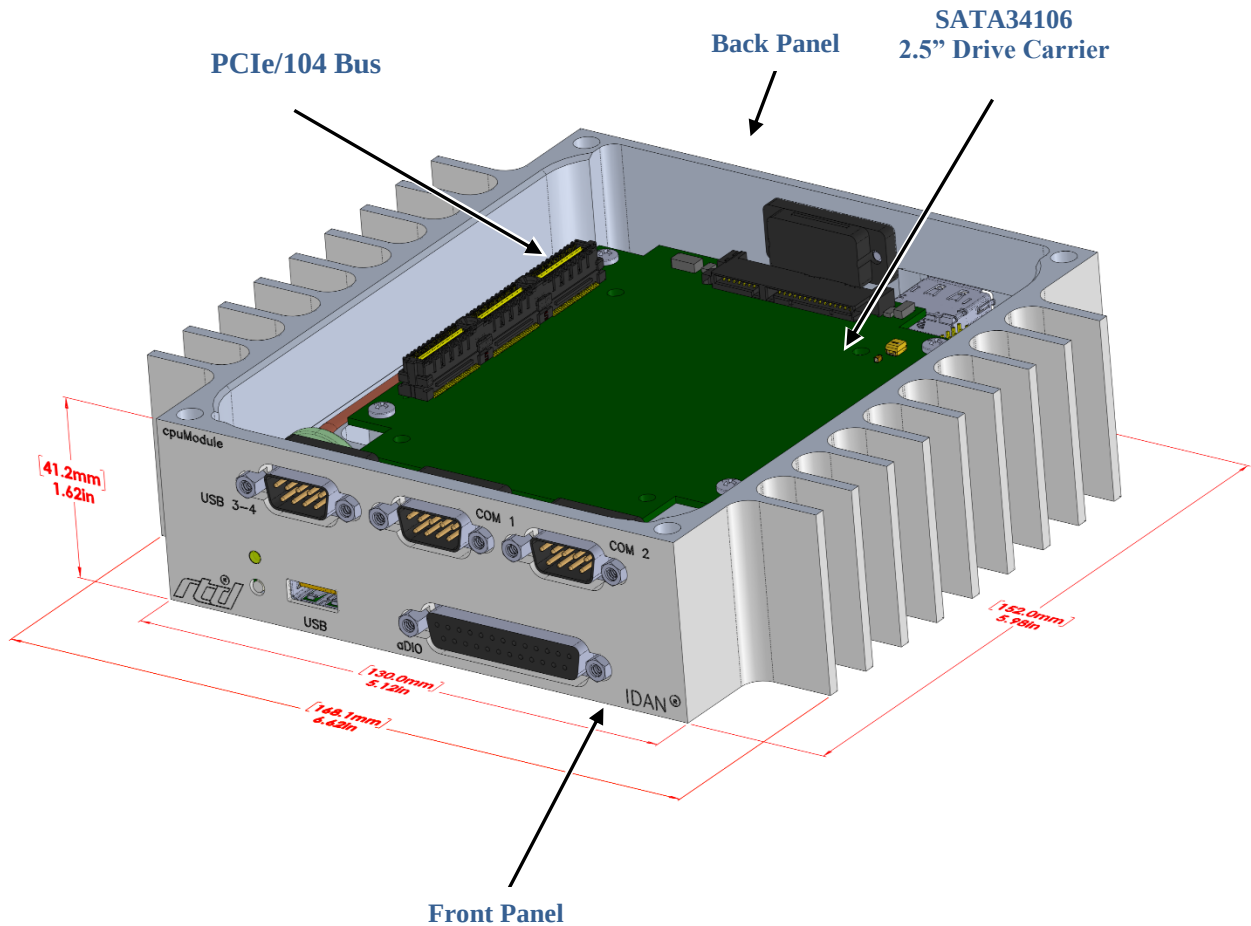
Also, inside the IDAN-CMA34KB unit is a CMOS battery which is connected to the battery input connection of the cpuModule's Utility Port 2.0 connector. The frame of the IDAN-CMA34KB brings out the cpuModule's multi-color LED as well as the cpuModule's reset button. While a power button is not provided on the exterior of the frame, the reset button is configurable as a power button through the BIOS Setup utility.

For additional flexibility, the 25-pin "D" connector which provides connections to the cpuModule's aDIO connector (**CN6**) includes connections to several pins on the Utility Port 2.0 connector, which include the reset button input, power button input, and RTC battery input.

For more information on cpuModule's battery input and reset button input on the Utility Port 2.0 connector, refer to the *Utility Port 2.0 Connector (CN5)* on page 39.

IDAN Dimensions

The IDAN frame for the CMA34KB is shown in the figure below:



IDAN Connector Locations

The diagrams below show the connector locations for the headers of the IDAN-CMA34KB as they are brought out on the front and back panels of the IDAN-CMA34KB. For a full description of each connector on the IDAN-CMA34KB, refer to *Connecting the cpuModule* on page 34. There is not an IDAN-CMX34KB at this time.

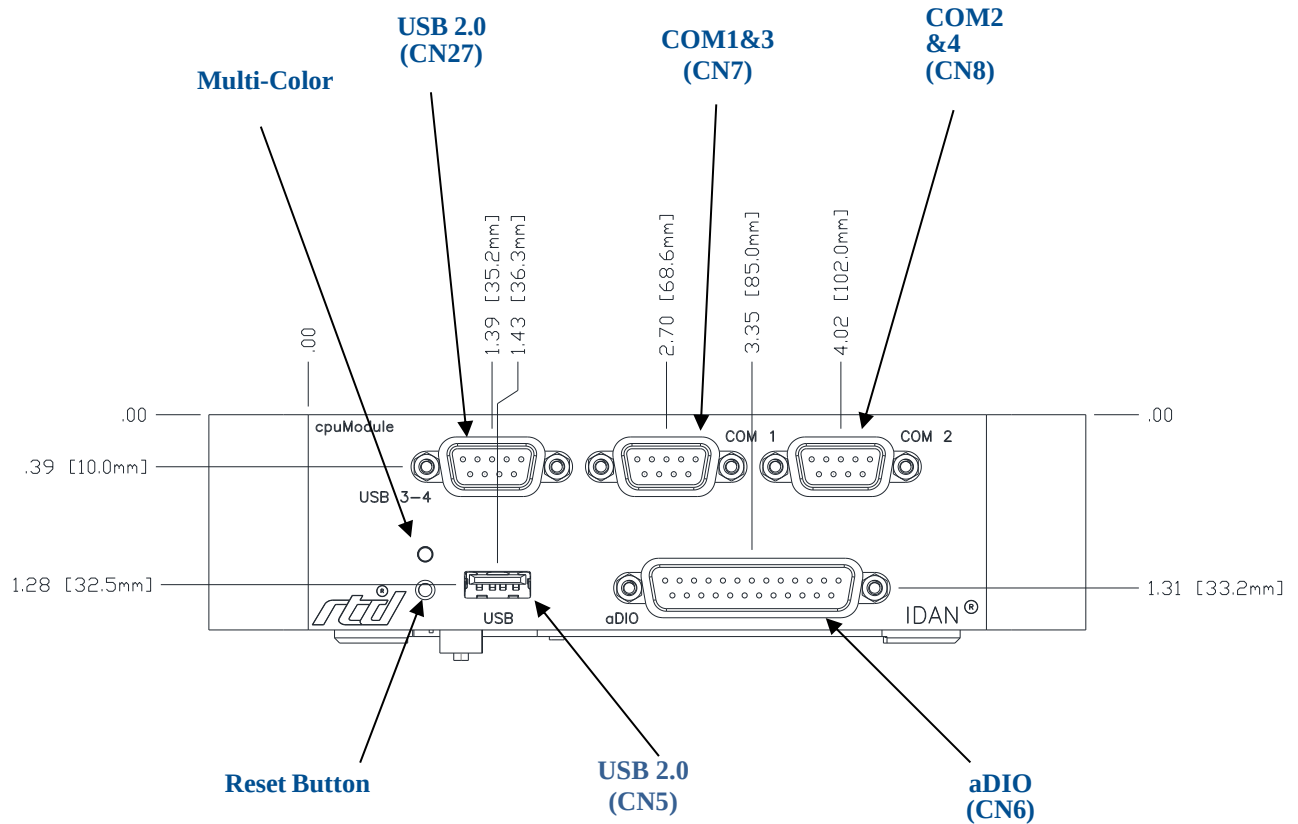


Figure 17 IDAN-CMA34KB Front Panel

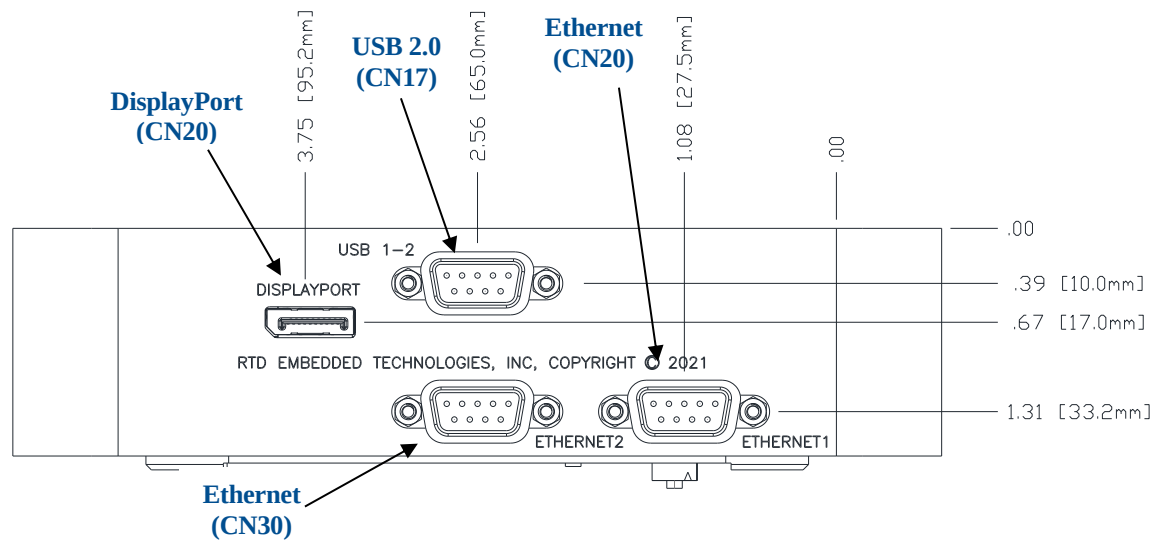


Table 59 IDAN-CMA34KB Connectors

CPU Designator	Function	Connector Description	IDAN Panel Connector	Mating Connector
CN4	DisplayPort	DisplayPort (female)	Molex 47272 series	Molex 68783 series
CN5	USB Port (Utility Port 2.0)	USB Type A	Series “A” Receptacle	Series “A” Plug
CN6	aDIO	25-pin D (female)	Adam Tech DB25SD	Adam Tech DB25PD
CN7	Serial Port 1 (COM1&3)	9-pin D (male)	Adam Tech DE09PD	Adam Tech DE09SD
CN8	Serial Port 2 (COM2&4)	9-pin D (male)	Adam Tech DE09PD	Adam Tech DE09SD
CN17	USB 2.0	9-pin D (male)	Adam Tech DE09PD	Adam Tech DE09SD
CN27	USB 2.0	9-pin D (male)	Adam Tech DE09PD	Adam Tech DE09SD
CN20 and CN30	Ethernet	9-pin D (male)	Adam Tech DE09PD	Adam Tech DE09SD

External I/O Connections

Table 60 COM1/COM2 (Single Port Mode)— 9-Pin D Connector (male)

CPU Pin	IDAN Pin (DB-9)	Single RS-232	Single RS-422	Single RS-485
1	1	DCD	TXD-	D-
2	6	DSR	—	—
3	2	RXD	TXD+	D+
4	7	RTS	—	—
5	3	TXD	RXD+	—
6	8	CTS	—	—
7	4	DTR	RXD-	—
8	9	RI	—	—
9	5	GND	GND	GND

Table 61 COM1/COM2 (Dual Port Mode)— 9-Pin D Connector (male)

CPU Pin	IDAN Pin (DB-9)	Dual RS-232	Dual RS-422	Dual RS-485
1	1	DCD1	TXD1-	D1-
3	2	RXD1	TXD1+	D1+
5	3	TXD1	RXD1+	—
7	4	RTS1	RXD1-	—
9	5	GND	GND	GND
2	6	CTS1	TXD2-	D2-
4	7	TXD2	TXD2+	D2+
6	8	RXD2	RXD2+	—
8	9	RI1	RXD2-	—

Table 62 aDIO — 25-Pin D Connector (female)

IDAN Pin #	aDIO Port	CPU Pin # (CN6)	CPU Pin # (CN5)
1	P0-0	1	—
2	P0-2	3	—
3	P0-4	5	—
4	P0-6	7	—
5	Strobe 0	9	—
6	P1-0	11	—
7	P1-2	13	—
8	GND	15	—
9	reserved	—	—
10	reserved	—	—
11	reserved	—	—
12	Soft Power Button Input	—	5
13	GND (for RTC Battery Input)	—	7
14	P0-1	2	—
15	P0-3	4	—
16	P0-5	6	—
17	P0-7	8	—
18	Strobe 1	10	—
19	P1-1	12	—
20	P1-3	14	—
21	+5 V	16	—
22	reserved	—	—
23	reserved	—	—
24	Push-Button Reset Input	—	3
25	RTC Battery Input	—	9

Table 63 DisplayPort Connector

Pin	Signal	Standard Function	DVI / HDMI mode ¹	In/Out
1	LN0+	Main Link, Lane 0 (positive)	TMDS Channel 2 (positive)	out
2	GND	Ground	Ground	out
3	LN0-	Main Link, Lane 0 (negative)	TMDS Channel 2 (negative)	out
4	LN1+	Main Link, Lane 1 (positive)	TMDS Channel 1 (positive)	out
5	GND	Ground	Ground	out
6	LN1-	Main Link, Lane 1 (negative)	TMDS Channel 1 (negative)	out
7	LN2+	Main Link, Lane 2 (positive)	TMDS Channel 0 (positive)	out
8	GND	Ground	Ground	out
9	LN2-	Main Link, Lane 2 (negative)	TMDS Channel 0 (negative)	out
10	LN3+	Main Link, Lane 3 (positive)	TMDS Clock (positive)	out
11	GND	Ground	Ground	out
12	LN3-	Main Link, Lane 3 (negative)	TMDS Clock (negative)	out
13	CFG1	Configuration Pin 1	Cable Adapter Detect	out
14	CFG2	Configuration Pin 2	Consumer Electronics Control ²	out
15	AUX+	Auxiliary Channel (positive)	DDC Clock	in/out
16	GND	Ground	Ground	out
17	AUX-	Auxiliary Channel (negative)	DDC Data	in/out
18	HPD	Hot Plug Detect	Hot Plug Detect	in
19	DPG	Return for DPV	Return for DPV	out
20	DPV	+3.3V DC Power	+3.3V DC Power	out

1. Requires special passive adapter

2.HDMI mode only

Facing DisplayPort connector, the pinout is:

19	17	15	13	11	9	7	5	3	1
DPG	AUX-	AUX+	CFG1	GND	LN2-	LN2+	GND	LN0-	LN0+
DPV	HPD	GND	CFG2	LN3-	LN3+	GND	LN1-	LN1+	GND
20	18	16	14	12	10	8	6	4	2

Table 64 Ethernet — 9-Pin D Connector (male)

IDAN Pin #	RJ-45 Pin	Signal	CPU Pin #
1	3	B+ (RX+)	1
2	4	C+	3
3	1	A+ (TX+)	5
4	7	D+	7
5	—	Ground	9
6	6	B- (RX-)	2
7	5	C-	4
8	2	A- (TX-)	6
9	8	D-	8

Table 65 USB 2.0 — 9-Pin D Connector (male)

IDAN Pin #	Signal	Function	Mode	CPU Pin #
1	VCC1	+5V to USB1	Output	1
2	Data USB1-	USB1 Data-	Input/Output	3
3	Data USB1+	USB1 Data+	Input/Output	5
4	GND	GND	—	7
5	Shield GND	Shield GND	—	9
6	VCC2	+5V to USB2	Output	2
7	Data USB2-	USB2 Data-	Input/Output	4
8	Data USB2+	USB2 Data+	Input/Output	6
9	GND	GND	—	8

Appendix D Additional Information

Application Notes

RTD offers many application notes that provide assistance with the unique feature set of the CMX34KB /CMA34KB cpuModule. For the latest application notes, refer to the RTD website.

Drivers and Example Programs

To obtain the latest versions of drivers and example programs for this cpuModule, refer to the RTD website.

Interrupt Programming

For more information about interrupts and writing interrupt service routines, refer to the following book:

Interrupt-Driven PC System Design
by Joseph McGivern
ISBN: 0929392507

Serial Port Programming

For more information about programming serial port UARTs, consult the following book:

Serial Communications Developer's Guide
by Mark Nielson
ISBN: 0764545701

PC/104 Specifications

A copy of the latest PC/104 specifications can be found on the webpage for the PC/104 Embedded Consortium:

<http://www.pc104.org>

Appendix E Limited Warranty

RTD Embedded Technologies, Inc. warrants the hardware and software products it manufactures and produces to be free from defects in materials and workmanship for one year following the date of shipment from RTD Embedded Technologies, Inc. This warranty is limited to the original purchaser of product and is not transferable.

During the one-year warranty period, RTD Embedded Technologies will repair or replace, at its option, any defective products, or parts at no additional charge, provided that the product is returned, shipping prepaid, to RTD Embedded Technologies. All replaced parts and products become the property of RTD Embedded Technologies. Before returning any product for repair, customers are required to contact the factory for a Return Material Authorization number.

This limited warranty does not extend to any products which have been damaged because of accident, misuse, abuse (such as: use of incorrect input voltages, improper or insufficient ventilation, failure to follow the operating instructions that are provided by RTD Embedded Technologies, “acts of god” or other contingencies beyond the control of RTD Embedded Technologies), or because of service or modification by anyone other than RTD Embedded Technologies. Except as expressly set forth above, no other warranties are expressed or implied, including, but not limited to, any implied warranties of merchantability and fitness for a particular purpose, and RTD Embedded Technologies expressly disclaims all warranties not stated herein. All implied warranties, including implied warranties for merchantability and fitness for a particular purpose, are limited to the duration of this warranty. In the event the product is not free from defects as warranted above, the purchaser's sole remedy shall be repair or replacement as provided above. Under no circumstances will RTD Embedded Technologies be liable to the purchaser or any user for any damages, including any incidental or consequential damages, expenses, lost profits, lost savings, or other damages arising out of the use or inability to use the product.

Some states do not allow the exclusion or limitation of incidental or consequential damages for consumer products, and some states do not allow limitations on how long an implied warranty lasts, so the above limitations or exclusions may not apply to you.

This warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

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